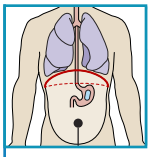
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LOWER BODY NEGATIVE PRESSURE: PHYSIOLOGICAL EFFECTS, APPLICATIONS, AND IMPLEMENTATION

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Goswami N, Blaber AP, Hinghofer-Szalkay H, Convertino VA. Lower Body Negative Pressure: Physiological Effects, Applications, and Implementation. *Physiol Rev* 99: 807–851, 2019. Published December 12, 2018; doi:10.1152/physrev.00006.2018.—This review presents lower body negative pressure (LBNP) as a unique tool to investigate the physiology of integrated systemic compensatory responses to altered hemodynamic patterns during conditions of central hypovolemia in humans. An early review published in *Physiological Reviews* over 40 yr ago (Wolthuis et al. *Physiol Rev* 54: 566–595, 1974) focused on the use of LBNP as a tool to study effects of central hypovolemia, while more than a decade ago a review appeared that focused on LBNP as a model of hemorrhagic shock (Cooke et al. *J Appl Physiol (1985)* 96: 1249–1261, 2004). Since then there has been a great deal of new research that has applied LBNP to investigate complex physiological responses to a variety of challenges including orthostasis, hemorrhage, and other important stressors seen in humans such as microgravity encountered during spaceflight. The LBNP stimulus has provided novel insights into the physiology underlying areas such as intolerance to reduced central blood volume, sex differences concerning blood pressure regulation, autonomic dysfunctions, adaptations to exercise training, and effects of space flight. Furthermore, approaching cardiovascular assessment using prediction models for orthostatic capacity in healthy populations, derived from LBNP tolerance protocols, has provided important insights into the mechanisms of orthostatic hypotension and central hypovolemia, especially in some patient populations as well as in healthy subjects. This review also presents a concise discussion of mathematical modeling regarding compensatory responses induced by LBNP. Given the diverse applications of LBNP, it is to be expected that new and innovative applications of LBNP will be developed to explore the complex physiological mechanisms that underline health and disease.

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I. INTRODUCTION

การใช้ความดันลบในส่วนล่างของร่างกาย (Lower Body Negative Pressure หรือ LBNP) เป็นวิธีการที่เฉพาะเจาะจง เนื่องจากมันทำให้เลือดเคลื่อนที่จากส่วนบน (เหนือไหล่ส่วนล่าง) ไปยังช่วงส่วนล่างของร่างกายโดยไม่มีผลกระทบจากการเปลี่ยนจากท่านอนไปยังท่าตั้งตรง ซึ่งอาจเกิดผลกระทบที่แตกต่างกัน เช่น การเปลี่ยนแปลงรูปแบบการกระตุ้นในเซ็นเซอร์โอโทลิธิกหรือการโหลดน้ำหนักตามท่าตั้งตรงทั้งหมดลงบนส่วนล่างของขาได้ เทคนิคนี้ได้ถูกนำมาใช้เป็นเครื่องมือทดลองเพื่อศึกษาการตอบสนองทางไฮโดไดแมนด์และเนโรคต่าง ๆ ที่เกิดขึ้นหลังจากการลดปริมาณของเลือดในส่วนกลางของร่างกายเป็นเวลาหลายทศวรรษแล้ว

การลดสภาพพร้อมท่านอนหรือการอยู่ในอวกาศเป็นหัวข้ออื่นที่ LBNP ได้ถูกใช้ทั้งเป็นเครื่องมือในการสำรวจและเครื่องมือในการฝึกทรีดทางหัวใจเสมอแล้ว

LBNP ลดการส่งกลับของเลือดไปที่หัวใจโดยทำให้เลือดสะสมที่ส่วนล่างของร่างกาย (โดยมีเส้นแบ่งระหว่าง "บน" และ "ล่าง" ขึ้นอยู่กับตำแหน่งซีล) การลดปริมาณเลือดในส่วนกลางนี้ทำให้เกิดภาวะขาดน้ำหนักในการทำงานของบารอเรฟล็กซ์ที่อยู่ในทางเลือดและทางหัวใจ-ปอด ซึ่งเริ่มต้นการกระตุ้นฮิโปเทนซินที่นำไปสู่การเพิ่มอัตราการเต้นหัวใจและการหดเส้นเลือด (1, 305) ซึ่งช่วยในการพยายามปรับความดันเพื่อรักษาการไหลเวียนเลือดทางแดนสมองและความดันเลือดที่เพียงพอ ภาพรวมของผลกระทบทางกลไกและสายระบายของ LBNP แสดงในรูปที่ 1 การหยุดใช้ LBNP ในทางตรงกันข้ามเหมือนกับผลของการทำท่าวัลซัลวา: เลือดเคลื่อนไปที่ท้องอกทำให้เกิดการเพิ่มความดันเลือดชั่วขณะและลดอัตราการเต้นหัวใจ

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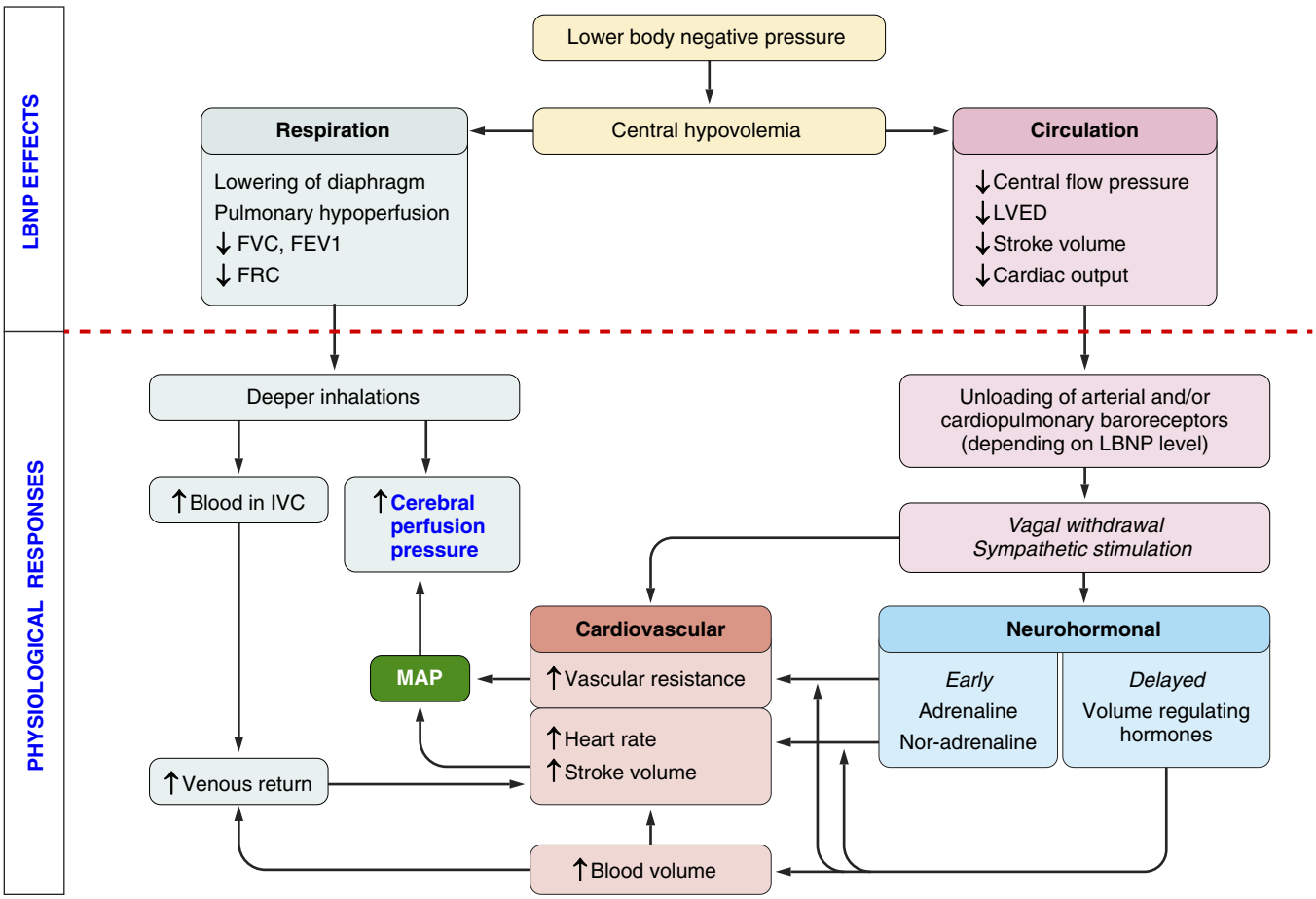


FIGURE 1. Overview of mechanical effects and physiological responses of lower body negative pressure (LBNP). FVC, forced vital capacity; FEV_1, forced expiratory volume in 1 s; FRC, functional residual capacity; LVED, left ventricular end-diastolic volume; IVC, inferior vena cava; MAP, mean arterial pressure.

BOX 1 outlines the role of LBNP as an important research tool and some of its translational and clinical applications.

A. Historical Perspectives

As early as 1834, subatmospheric pressure was used for producing dry operating fields during surgery to remove blood from diseased organs as reported by Junod (396). **TABLE 1** shows the historical development of LBNP as a research tool (summarized from Ref. 36).

B. Applications of LBNP

In addition to being a tool for basic physiological research, LBNP has proven valuable for several clinical and spaceflight applications. Some of these include the following:

1. As an assessment tool for autonomic system function in patients (290);
2. To increase footward loading for resistance training, an aspect that could have application during orthopedic rehabilitation (160), −100 mmHg LBNP causes a footward force equal to one body weight (383);
3. For counteracting balance and postural modifications induced by bedrest confinement (91);
4. As a convenient test for screening pilots, as LBNP is less stressful compared with centrifugation exposure (55);
5. To partially reverse the cephalad fluid shifts that occur in weightlessness. LBNP is used along with "salt-soak" in the Russian space program (122);
6. LBNP combined with exercise has been used to simultaneously stress the cardiovascular and the musculoskeletal systems (268);
7. For studying the influence of environmental stressors on physiological responses [e.g., effects of whole-body heating during LBNP, which simulates the stressors to which military personnel are exposed during standing in hot climates (80)];
8. For investigating the effects of acute hemorrhagic shock (78, 176);
9. As a countermeasure to prevent cardiovascular deconditioning and to reduce orthostatic intolerance, which can arise following bedrest confinement or on return to Earth from the microgravity environment of spaceflight (126, 128, 161, 179, 220, 385);

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BOX 1. Role of LBNP as an important research tool

- LBNP provides a noninvasive method capable of inducing progressive reductions in central blood volume in conscious humans that can model aspects of the hemodynamics associated with bleeding. This is particularly important in emergency or military medicine.
- LBNP application in seated subjects (adjusted to maintain left atrial diameter) has been used to understand the relative contributions of low and high pressure receptors on arginine vasopressin release. This knowledge is important for heart failure patients (who already have higher levels of vasoconstrictor hormones and increased systemic vascular resistance) in whom better sodium and fluid regimes/therapies could be developed (see Ref. 302).
- LBNP has been used to assess the effect of the serotonin reuptake inhibitors on cardiovascular reflexes (357).
- LBNP, in combination with treadmill exercise, has been used to provide gravity-like conditions to load the musculoskeletal and cardiovascular systems in space. On Earth, this combined procedure not only protects persons subjected to head-down tilt bed rest against orthostatic deconditioning but also improves muscular strength, spine function, and bone metabolism (347).
- LBNP, in combination with head up tilt, has been used to assess effects of Traditional Chinese Medicine (121), mental challenge (139), and application of artificial gravity on the time taken to develop presyncope (that is, orthostatic tolerance) (105, 135).

10. To study the effects of reduced gravity environments [e.g., as found on the Moon or Mars (332)];
11. As a substitute to study the effects of varying G loads (as in centrifugation), since physiological responses (e.g., hypotension and tachycardia) are similar (142, 239, 375);
12. As a tool to compensate for head down tilt-induced hormonal, thoracic impedance, transvascular fluid shift, and hemodynamic effects (169, 276);
13. As a tool to evaluate the effects of central volume expansion versus hemodilution on renal excretions of sodium following an intravenous saline infusion (191);
14. As a tool to understand the effects of narrowing (decreasing) arterial pulse pressure on the LBNP-induced heart rate response (276);
15. As a tool to assess which type of baroreceptors govern vasopressin and renin releases during posture changes (276, 302);
16. As a technique for studying cardiovascular compensatory mechanisms during aerobic exercise and whether adaptation to aerobic exercise training influences tolerance during central hypovolemia (70, 229);
17. As a tool for calculating lower limb venous compliance. Using special strain gauges placed around the calf muscle, graded LBNP-induced alterations in lower leg volume can be measured (detailed in Ref. 55). The lower limb venous compliance can be measured by the ratio of the estimated change in calf volume to the change in LBNP pressure;
18. As a countermeasure for the increased cerebrospinal fluid on Earth or in space, as it directly influences cerebrospinal fluid surrounding the optic nerve (254) (detailed in sect. V*E*).

The LBNP apparatus can be mounted on a tilt table that allows for experimental protocols to compare mechanistic responses to orthostatically isolated footward (head up tilt, HUT) and headward (head down tilt, HDT) fluid shifts (291).

This review provides a detailed overview of LBNP-induced physiological responses and compares LBNP with its experimental advantages to other models that are currently used to simulate central hypovolemia and orthostatic challenge. It contains an inclusive collection of evidence from the scientific literature, aimed at providing novel insights into the physiology related to important areas such as tolerance to central hypovolemia, sex differences in blood pressure regulation, autonomic dysfunctions, exercise training, and hemorrhagic shock as well as mechanisms underlying these responses and mathematical modeling of physiological control systems.

II. OVERVIEW OF PHYSIOLOGICAL EFFECTS INDUCED BY LBNP

A. Specific Effects of LBNP on Different Physiological Systems

1. Cardiovascular system

LBNP elicits reproducible reflexive hemodynamic and cardiovascular control responses (72, 329), similar to those following increased G load (e.g., increases in heart rate and total peripheral resistance) (16), which ensure a compensatory effort to maintain blood pressure and cerebral perfusion (27).

LBNP is commonly used as a means to study cardiovascular reflex responses to central hypovolemia (78, 396). Typical cardiovascular effects induced by LBNP are best illustrated by the classic data published more than three decades ago by Blomqvist and Stone (27, 193) and presented here as **FIGURE 2**.

Two distinct cardiovascular effects accompany LBNP: *1*) biophysical impacts of reduced central circulating blood volume on pressure-flow relationships, which in turn initiate *2*) various compensatory effector responses mediated by neuroendocrine reflex mechanisms (**FIGURES 1 and 2**). As to the first effect, chamber decompression causes blood to move from the central circulation to the lower body below

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Table 1. *Historical development of LBNP applications on Earth and in Space and reported cardinal effects of LBNP*

Timeline	Researcher	Observations Related to LBNP Application
1962–1963	Dr. Duane Gavelin	Reported to have built the first prototype of the LBNP device for spaceflight-related research. Observed that −30 mmHg LBNP induced cardiovascular responses as upright standing in 1 g.
1963	Greenfield et al.	Presented the effects of sudden overload on the cardiovascular system by release of the LBNP.
1965–1969	Stevens and Lamb (353)	Further development of the LBNP prototype built by Dr. Gavelin for assessing orthostatic intolerance.
		LBNP proposed to have advantages over tilt table testing, as LBNP did not affect cerebral blood flow and thus did not affect baroreceptor responses and the venous tone.
		No saddle support built into the device.
		Application of LBNP in bed-rested subjects improved orthostatic tolerance.
1969	Musgrave et al. (269)	Changes in lower leg volume during −40 mmHg LBNP similar to those during standing up.
		Nonstimulation of carotid baroreceptors during LBNP means that greater LBNP levels required to elicit same heart responses as during upright standing.
1971	Musgrave et al. (270)	70° head-up tilt-induced responses are similar to those of −40 mmHg LBNP.
1971	Gazenko et al. (126)	In-flight LBNP used (Vetter). This was followed by the development of the Chibis-LBNP suit.
1974	Berry	In the Apollo program, LBNP was used pre- and post-spaceflight to test for orthostatic intolerance.
1977	Johnson et al. (194)	In the Skylab program, LBNP was used pre, during, and post spaceflight.
		LBNP applied during the flight acted as a good predictor of post-spaceflight orthostatic intolerance.
		Astronauts adapted to inflight-LBNP induced orthostatic intolerance after 4 wk in space.
1989–1995	Charles and Lathers (216)	LBNP is used in-flight to alleviate post-spaceflight orthostatic intolerance.
1998	Ertl et al. (100)	First direct recordings of muscle sympathetic nerve activity (MSNA) during LBNP were made in space during the *Neurolab* mission. The German space agency developed a flexible LBNP chamber with zippers on the side for access to the peroneal nerve.

Summarized from Campbell and Charles (36). For details of the references, please refer to Ref. 36.

the iliac crest as evidenced by a linear increase in leg volume. This translates into reduced cardiac filling pressure, reduced left ventricular end-diastolic volume (LVED), and consequently decreases in stroke volume and cardiac output by the Frank-Starling mechanism. Baroreflex feedback and neuroendocrine compensatory responses are activated to protect tissue oxygenation through maintenance of mean arterial pressure.

The onset of relative hypotension secondary to central hypovolemia marks the threshold when falling arterial pressure begins to fail to maintain adequate tissue perfusion (109, 381). Barcroft et al. (12) eloquently demonstrated more than 70 yr ago that both pronounced decreases in heart rate and vascular resistance contributed to an immediate drop in systolic blood pressure (SBP) from 120 to <40 mmHg in a human volunteer during a controlled hemorrhage of 1080 ml. However, more recent data obtained from a protocol of progressive LBNP demonstrated that the significant hypotension associated with presyncope is marked by distinct time courses of two mechanistic phases (331). The initial phase is marked by a reduced peripheral vascular resistance. Despite this vasodilation, mean arterial pressure is maintained at a level (>80 mmHg) commensurate with adequate tissue (e.g., brain) perfusion and oxygenation (109). However, it is only after a delay of minutes following the initial drop in blood pressure that the second phase of hypotension occurs with a rapid reduction in mean arterial pressure to <60 mmHg associated with a pronounced bradycardia of ≥30 beats/min reduction in heart rate (232, 331). This level of perfusion pressure results in unsustainable cerebral tissue oxygenation, as reflected by the onset of presyncopal symptoms within seconds following the bradycardia-induced hypotension. As such, it has been proposed that bradycardia represents a "last ditch effort" to save the organism by optimizing cardiac filling time and reducing cardiac work in critically low circulating volume states such as during high levels of LBNP when coronary blood flow is compromised (336).

2. Cerebral blood flow and LBNP

A) CEREBRAL BLOOD FLOW REGULATION AND LBNP. The concept that cerebral blood flow (CBF) is "autoregulated" was first gen-

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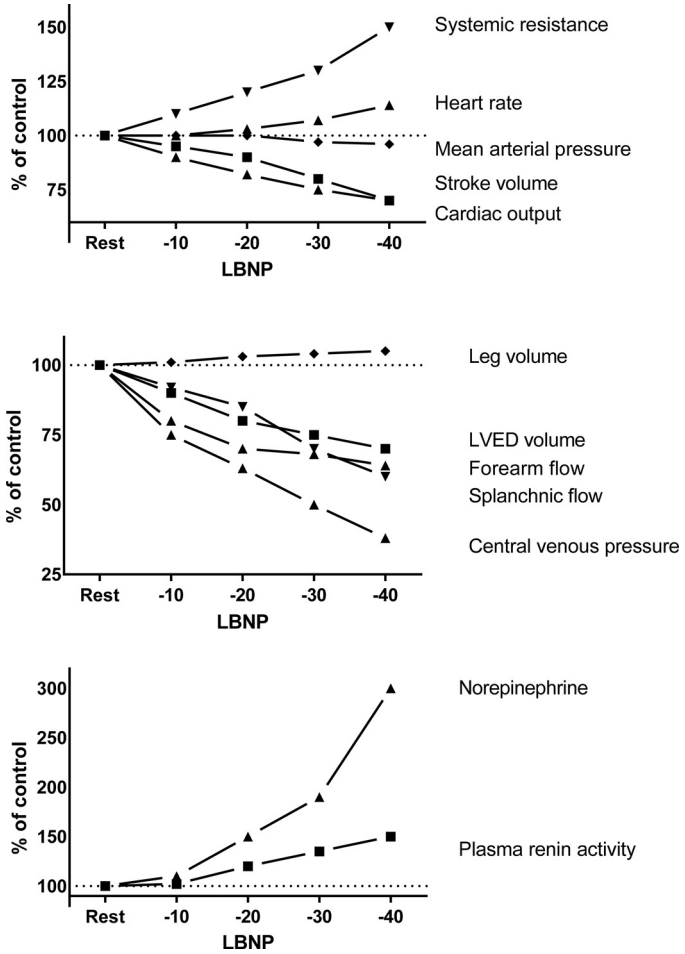


FIGURE 2. Hemodynamic and (selected) hormonal responses are induced by application of ramp lower body negative pressure (LBNP; −20 to −40 mmHg). [Figure drawn from data in Blomqvist and Stone (27) and Johnson et al. (193).]

erated more than 55 yr ago based on CBF measurements obtained from independent patient populations (213). If cerebral autoregulation existed in the purest form of maintaining blood flow across a wide range of reduced blood pressure, CBF might be expected to remain constant during progressive increases in LBNP until the time of presyncope when mean arterial pressure falls below 60 mmHg. Contrary to this paradigm, measures of middle cerebral artery blood velocity ($MCAv_{mean}$) during progressive LBNP **(FIGURE 3*A*)** support the well-documented observation of a reduction in CBF (11, 28, 94, 148, 158, 203, 218, 219, 227, 228, 236, 369, 370, 392, 402). However, there is evidence that the reduced CBF during LBNP is not homogeneous, with ~25% reductions in cortical gray matter blood volume with little change in white matter or basal ganglia regions (392).

The reduction in $MCAv_{mean}$ during LBNP is related to the lowering of regional cerebral O_2 saturation (148). However, 10–15% reductions in $MCAv_{mean}$ and regional cerebral oxygen saturation during LBNP were found to be less than expected given ~30% reduction in cardiac output (148, 225). Interestingly, pharmacological reductions >30% in CBF (measured by $MCAv_{mean}$) demonstrated that the brain extracts more oxygen in the face of cerebral hypoperfusion as a compensatory mechanism to buffer the impact of reduced oxygen delivery (227). These observations suggest the preservation of some degree of cerebrovascular regulation that functions to protect adequate cerebral oxygenation in normal healthy individuals despite systemic hypotension during progressive central hypovolemia induced by LBNP.

B) COGNITIVE FUNCTION RELATED TO CBF RESPONSES TO LBNP. Due to cerebrovascular regulation, presyncopal symptoms are delayed despite a reduction in CBF (314). When tests of cognitive function are conducted, the lower $MCAv_{mean}$ at −50 mmHg LBNP results in increased latency and reduced amplitude of event-related EEG potentials that are associated with slower reaction time and greater tracking error (130, 158). These results suggest that the reduction in cerebral perfusion induced by higher levels of LBNP translates to a threshold of lower cerebral tissue oxygenation sufficient to impair neurocognitive function.

C) CBF RELATIONSHIP WITH VENTILATION DURING LBNP. The hypocapnia resulting from hyperventilation and lower end-tidal carbon dioxide ($P_{ET_{CO_2}}$) during LBNP is associated with a reduction in cerebral vascular conductance (CVCi) and $MCAv_{mean}$ (218). The reduced cerebral perfusion with hyperventilation and lower $P_{ET_{CO_2}}$ does not appear to be associated with reduced cerebrovascular CO_2 responsiveness as indicated by no change in the slopes of the linear relationship between $P_{ET_{CO_2}}$ and $MCAv_{mean}$ (223).

D) CBF AND PRESYNCOPAL RESPONSES TO LBNP. Syncope is a sudden, temporary, and transient loss of consciousness, which is often accompanied by a loss in postural muscle tone (200). Presyncopal symptoms and signs include dizziness, hypotension, and sudden heart rate decrease. Syncope is discussed in detail in a separate section of this review, but here we provide details of the effects of LBNP on CBF that can lead to presyncopal responses.

A rapid fall in cerebral tissue oxygenation during the final seconds of LBNP before presyncope is associated with hemodynamic decompensation and hypotension (28, 130, 210, 331). Modeling of the effects of LBNP on cerebral hemodynamics revealed that the magnitude of the relationship between arterial blood pressure and $MCAv_{mean}$ increased at LBNP levels above −30 mmHg, suggesting onset and continued impairment to cerebrovascular regulation, while reduction in the $P_{ET_{CO_2}}$-$MCAv_{mean}$ relationship implied an attenuation of cerebral vasomotor reactivity (262). This model provides support for the concept that a progressive decrease in CBF during LBNP reflects a lower "cerebrovascular reserve to compensate for the increasingly unstable systemic circulation . . . that could ultimately lead to

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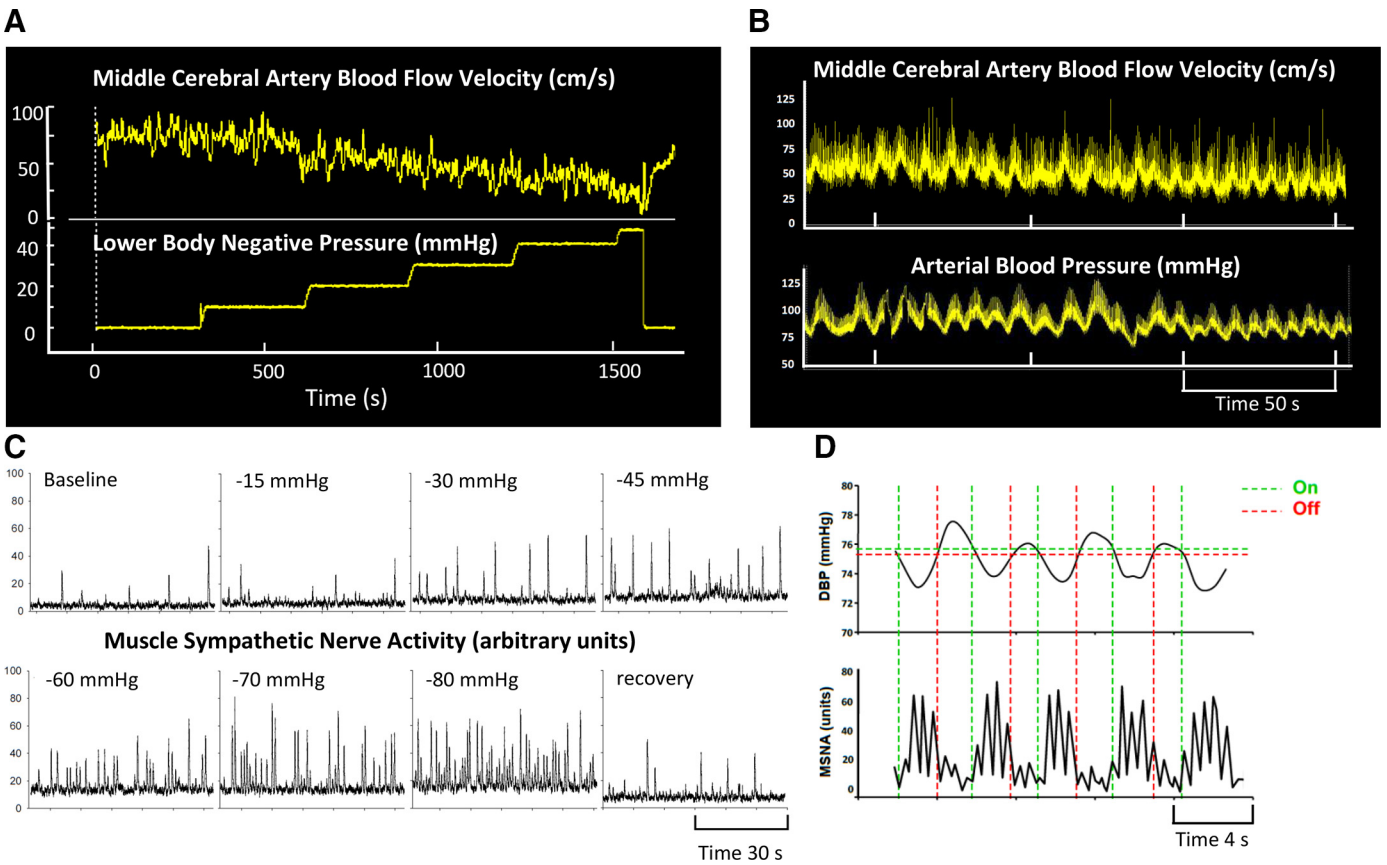


FIGURE 3. *A–D*: cerebral autoregulatory, blood pressure, and sympathetic nervous system responses to central hypovolemia induced by lower body negative pressure (LBNP). Although each component of the figure is from separate experiments and separate individual subjects (51, 67, 77, 314), the data show physiological consistency. *A*: effects of progressive LBNP on cerebral blood flow. Although autoregulation should maintain cerebral blood flow (CBF) in the face of decreasing blood pressure, in some persons there is a gradual decline in CBF. This may have contributed to presyncope at −50 mmHg in this subject. [From Rickards et al. (314).] *B*: the oscillations of CBF observed in *A* are presented in finer detail in a subject exhibiting high orthostatic tolerance at −80 mmHg. These oscillations exhibit the relationship between CBF (top tracing) and systemic arterial blood pressure variation (bottom tracing). As blood pressure oscillates, similar changes in CBF are seen. This information has been used by many researchers to assess the response characteristics of the cerebral vessels. [From Convertino et al. (67).] *C*: representative recordings of muscle sympathetic nerve activity (MSNA) obtained from a subject during progressive LBNP. With increasing LBNP, a clear increase in the strength and frequency of MSNA bursts is seen, which leads to increased vasoconstriction thus preventing reductions in blood pressure. In this subject, presyncope did not occur until −80 mmHg. [From Cooke et al. (77).] *D*: the relationship (i.e., coherence) between blood pressure oscillations and MSNA as a compensatory response recorded from a healthy human subject undergoing −60 mmHg LBNP. Variations in diastolic blood pressure (DBP) should trigger sympathetic autonomic vasoconstrictor responses. Direct measurement of MSNA allows for the monitoring of sympathetic responses that alter arteriolar diameter to sustain mean arterial pressure adequate to support sufficient CBF for prevention of presyncope. Data presented here show MSNA in response to rhythmic changes in DBP. Bursts of MSNA occur synchronized with a nadir in DBP with quiescent phases during peaks. Further details related to *D* are provided in sect. II*A4*B. [From Convertino (51).]

cerebral hypoperfusion and syncope" (262). If low cerebral perfusion represents a compromise to a primary compensatory mechanism for maintaining adequate cerebral oxygenation, then increasing $MCAv_{mean}$ might be expected to delay the onset of presyncopal symptoms. Contrary to this notion, a 25% higher $MCAv_{mean}$ resulting from breathing a hypercapnic gas mixture (5% CO_2) failed to enhance LBNP tolerance (236). These experimental results indicate that reduced cerebral perfusion associated with hypocapnic-induced cerebral vasoconstriction alone cannot explain the mechanism underlying the onset of LBNP-induced presyncope.

There is now compelling evidence that oscillations in CBF, like those observed in systemic blood pressure (**FIGURE 3*B***), represent an important mechanism for cerebrovascular regulation. **FIGURE 3*B***, *top panel*, shows a representative tracing of $MCAv_{mean}$ from a subject undergoing −80 mmHg LBNP. Similar to the arterial pressure recordings (**FIGURE 3*B***, *bottom panel*), CBF demonstrates an obvious and con-

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tinuous oscillatory pattern. Interestingly, these patterns of oscillation may be more important to cerebrovascular regulation than absolute velocity as evidenced by higher LBNP tolerance being associated with elevated oscillations despite similar $MCAv_{mean}$ levels (315). These data speak to the physiological importance of using LBNP as an experimental platform to measure dynamic relationships such as oscillations in pressure and flow that accurately reflect the physiological status of cerebral circulation rather than relying on average responses. Reduced MCAv pulsatility during LBNP may also contribute to the development of imminent syncope (285, 361).

3. Effects of LBNP on pulmonary function

The application of LBNP causes a shift of the diaphragm in the caudal direction, pulmonary hypoperfusion, and increase in forced vital capacity (FVC), forced expiratory volume in 1 s ($FEV_{1.0}$), and functional residual capacity without alterations in pulmonary compliance (47, 167). These functional changes are explained by a reduced thoracic blood volume due to the translocation of fluid from the thorax to the lower body, subsequently allowing for more air to be moved faster without the impedance caused by circulating pulmonary blood volume. The notion that blood in the lung reduces FVC and $FEV_{1.0}$ is further supported by the relationship of larger increases in FVC and $FEV_{1.0}$ during LBNP in subjects with larger lung volumes that allow a greater absolute shift in blood to or from the thorax (47). Finally, the absence of change in lung compliance provides evidence that the increases in FVC and $FEV_{1.0}$ during LBNP result from changes in thoracic blood volume rather than mechanical properties of the lungs themselves.

Diffusion of the lung measured by the carbon monoxide technique (DL_{CO}) and pulmonary capillary blood volume decrease linearly with each increasing stage of LBNP (140, 400). The reduction in DL_{CO} appears to be due primarily to the result of reduced pulmonary capillary blood volume since the diffusion capacity of the pulmonary membrane is unaffected (140, 400).

Observations reported from experiments in which breath-by-breath measurements of pulmonary capillary oxygen transfer and ventilation were made during progressive levels of LBNP revealed that loss in blood oxygen stores in the lung was directly related to the intensity of LBNP (235). The peak rise in pulmonary capillary oxygen transfer after a release of LBNP was always preceded by a decrease in leg volume, indicating that changes in oxygen stores were related to blood volume shifts. The return of oxygen-depleted pooled blood to the central circulation during the first minute of recovery caused significant hyperpnea. Three compartment lung model analyses from alveolar and arterial blood samples at −60 mmHg LBNP showed a 90% increase in the alveolar dead space fraction with a slight decline in the effective compartment. The less effective lung perfusion during LBNP may explain a 30% increase in ventilation equivalent for oxygen (235).

In a study in which eight subjects conducted supine dynamic leg exercise (0 to 100 W) with and without exposure to −50 mmHg LBNP, Eiken et al. (98) showed an altered respiratory drive. Steady-state heart rate was higher in LBNP, and inspired minute ventilation and drive to breathe (measured as the occluded inspiratory pressure at 0.1, $P_{0.1}$) were significantly lower than the control condition. The reduction in the drive to breathe during LBNP at similar work rates to the control (no LBNP) is consistent with reduced chemoreceptor stimulation, most likely due to improved blood flow in the exercising leg muscles. This was further supported by an investigation of oxygen uptake kinetics (183). Although there was no difference in peak oxygen uptake ($\dot{V}O_2$) and ventilatory threshold between the two supine conditions (with and without LBNP), the rate at which the subjects reached steady state after a transition to a new work load was faster with LBNP, consistent with improved muscle perfusion in the legs.

A) EFFECTS OF LBNP-INDUCED VENTILATION CHANGES ON HEMODYNAMICS. Although pulmonary function is usually perceived as a fundamental mechanism for gas exchange, acid-base balance, endocrine release, and immune responses, its role in the control of venous return and cardiac filling resulting from the negative intrathoracic pressure created during inspiration is less appreciated (266). Ventilatory volume ($\dot{V}E$) changes very little during LBNP until a level that represents >80% of LBNP tolerance, at which point $\dot{V}E$ increases >50% primarily as a result of elevated tidal volume (65). Interestingly, a reduction in $P_{ET_{CO_2}}$ occurs well in advance of increased $\dot{V}E$ at ~40% of LBNP tolerance (257). The deeper inhalations associated with greater tidal volume employs the so-called "respiratory pump" to result in greater reductions in intrathoracic pressure (i.e., a greater vacuum) that is translated to elevate blood flow in the inferior vena cava and improve venous return (266). The importance of the "respiratory vacuum pump" on supporting stable hemodynamics during LBNP has been eloquently demonstrated by increased LBNP tolerance associated with protection against reductions in cardiac output (325), blood pressure (62), and cerebral perfusion (314).

In addition to increased ventilation associated with reductions in central blood volume induced by LBNP in humans (65, 186), tissue perfusion is reduced with sympathetically mediated vasoconstriction (57, 71), resulting in lower tissue oxygen content (349, 350, 381). However, the elevated respiratory response to LBNP represents a hyperventilation in response to lower central blood volume rather than an increased requirement for gas exchange since end-tidal CO_2 is reduced (257) in the absence of changes in oxygen uptake, carbon dioxide production, arterial oxygen saturation, or blood lactate (65). As such, the endogenous "respiratory

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pump" may be a protective strategy characterized by faster and/or deeper breathing to optimize cardiac filling with central hypovolemic hypotension caused by LBNP.

In addition to its influences on central hemodynamics, the respiratory pump acts as a compensatory mechanism to assist with sustaining optimal CBF and oxygenation during LBNP. Reduced intrathoracic pressure caused by deeper inspirations during LBNP is transferred to the brain as reductions in intracranial pressure via an anatomical paraventricular plexus. Consequently, the pressure gradient for cerebral perfusion (i.e., cerebral perfusion pressure) is increased by lowering the intracranial pressure (i.e., resistant to flow) coincident with an elevated arterial pressure (69). This relationship between breathing and cerebral perfusion has been eloquently demonstrated in experiments where CBF is increased by regulating lower intrathoracic pressure with the application of resistance during inspiration (69).

Finally, experiments designed to investigate the relationship between pulmonary responses and hemodynamics support the notion that the pattern of flow represents a critical influence of respiration during LBNP. This concept is supported by the observation that greater reductions in intrathoracic pressure coincident with deeper inspirations during LBNP are associated with greater CBF oscillations, delayed expression of syncope symptoms (i.e., inadequate cerebral oxygenation), and increased tolerance to LBNP (313).

4. Autonomic system and LBNP

Compensatory responses associated with the control of cardiac output and peripheral vascular resistance during the reduction in central blood volume and hypotension induced by LBNP are under the control of parasympathetic (cardiac vagal) and sympathetic activity. These compensatory mechanisms work in concert through autonomically mediated baroreflexes to maintain global tissue oxygenation through the regulation of systemic (arterial) perfusion pressure (316, 321, 377).

A) CARDIAC BAROREFLEX RESPONSES DURING LBNP. The decrease in central blood volume caused by LBNP translates to a transient lowering of systemic flow and perfusion pressure (**FIGURES 1 AND 2**). The consequent reduction in arterial blood pressure provides a stimulus to baroreceptors located in both the aorta and carotid arteries that initiate vagal withdrawal to increase heart rate (252). Since the carotid-cardiac baroreflex has been attributed to primarily parasympathetic nerve traffic (112, 300), acute vagal cardiac withdrawal has been demonstrated during LBNP by progressive reductions in the calculated beat-to-beat spontaneous cardiac baroreflex sensitivity (66, 75) and frequency-derived components of heart period variability (75, 76, 300).

The interpretation of the contribution of a reflex heart rate response mediated through parasympathetic control may be best considered in the framework of time to respond to the hypotensive stimulus introduced by LBNP. Responses of multisynaptic sympathetically mediated reflexes require 2–3 s to occur (96), with maximum response requiring 5–10 s (321) in contrast to a rapid activation of the vagally mediated carotid-cardiac baroreflex which requires only milliseconds for a heart rate response (96). As such, it is likely that the response to minimize the reduction in cardiac output and adequate tissue perfusion pressure during the initial moments of a LBNP challenge is dependent on an important contribution of rapid cardiac parasympathetic withdrawal. This is supported by the observation that the vagally mediated carotid-cardiac baroreflex was the only reflex represented in a model for predicting the time of presyncope onset in a rapid LBNP profile (238, 239).

In addition to cardiac vagal withdrawal, unloading of various receptor populations (e.g., baroreceptors, chemoreceptors, muscle metaboreceptors) initiates a linear increase in efferent sympathetic traffic to the heart and peripheral vasculature with progressively increasing levels of LBNP (37, 59, 62, 77, 119, 195, 198, 309, 311, 319, 326, 330, 331, 334, 379, 391) as illustrated by the sympathetic nerve recordings in **FIGURE 3*C***. In the absence of vagally mediated cardiac control created by the use of atropine, there remains an equal magnitude of tachycardia during progressive LBNP compared with a placebo condition (57, 71). The results from these cholinergic blockade experiments demonstrate that elevated heart rate during LBNP-induced central hypovolemia is not solely dependent on parasympathetic withdrawal, but that the sympathetic limb of cardiac control represents a significant redundancy to maintain an appropriate reflex tachycardia during central hypovolemia. Furthermore, when the elevation in heart rate during LBNP is attenuated with beta-adrenergic blockade (i.e., propranolol), tolerance to central hypovolemia is dramatically reduced (57, 71). Taken together, these experiments demonstrate that sympathetic nerve activity to the heart plays a significant role in the tachycardia response to LBNP. As such, a compelling argument can be made that the autonomic nervous system influence on the heart represents a prominent mechanism underlying the elevation of heart rate for attenuating a drop in cardiac output during LBNP.

B) VASCULAR BAROREFLEX RESPONSES DURING LBNP. In addition to a subtle lowering of arterial blood pressure, reduced venous return during LBNP is reflected in a decrease in central venous pressure (CVP; **FIGURES 1 AND 2**). Lowered CVP during LBNP has been shown to be a stimulus to activate sympathetically mediated vasoconstriction (17, 120, 193, 246–248, 362). Prior to 1975, it was generally accepted that low levels of LBNP (−20 mmHg or less) exclusively unload the cardiopulmonary baroreceptors (124, 193, 316, 403). However, data from subsequent investigations con-

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ducted after 1985 indicated that carotid baroreceptors are also unloaded since small but consistent linear increases in heart rate were observed during LBNP levels of −15 and −20 mmHg (256, 293, 362, 365, 377). The latter experiments provide compelling evidence that sympathetically mediated vasoconstriction initiated by activation of both cardiopulmonary and carotid baroreceptor unloading are responsible for the increase in peripheral vascular resistance during exposure to LBNP.

Perhaps more important than simply maintaining arterial pressure during LBNP-induced central hypovolemia are the oscillatory pressure and flow patterns previously reported during reductions in central blood volume more than 65 yr ago (150). **FIGURE 3*B*** demonstrates that these oscillatory patterns exist in both systemic and cerebral circulations, particularly at high levels (−80 mmHg) of LBNP (314). The significance of blood pressure and flow oscillations as a compensatory response during LBNP is best illustrated by the two tracings presented in **FIGURE 3*D*** that were obtained from a healthy human subject undergoing −60 mmHg LBNP. The *top panel* represents the continuous beat-to-beat measurement of diastolic blood pressure which oscillated between 73 and 78 mmHg. Oscillatory blood pressure patterns represent a highly sensitive feedback mechanism that exists between the pressure stimulus to baroreceptors and the sympathetically mediated reflex nerve response that activates vascular smooth muscle constriction (66, 331). The *bottom panel* represents direct muscle sympathetic nerve activity (MSNA) recordings obtained from a subject's peroneal nerve. After reaching a specific pressure threshold (horizontal green broken line), descending pressure initiates an "on" stimulus (vertical green broken lines) that translates to a precise increase in sympathetic nerve activity. The subsequent arterial vasoconstriction results in increased vascular resistance and a compensatory elevation in blood pressure to support cerebral perfusion. The return of pressure to a specific threshold with ascending pressures (horizontal red broken line) initiates an "off" stimulus (vertical red broken lines) that results in a reflex-mediated reduction in sympathetic nerve activity and vasodilation to support perfusion of peripheral tissue. This tight oscillatory coupling between arterial blood pressure (stimulus) and sympathetic nerve activity (response) is an excellent example of the eloquent feedback system critical for the tight regulation of systemic blood flow patterns (i.e., small difference between the green and red horizontal broken lines) required for optimal perfusion to all tissues by altering blood flow between the brain and peripheral organs. Furthermore, the sensitivity of this intricate feedback mechanism increases during progressive LBNP-induced central hypovolemia as reflected by an elevated coherence between arterial pressure and sympathetic nerve activity (67, 77). Laurin et al. (217) quantified these interaction bursts at 0.1 Hz (Mayer waves) between MSNA and blood pressure changes during progressive steps of LBNP up to −60 mmHg in young, healthy subjects. Two new parameters for assessing the sympathetic nervous interaction with diastolic blood pressure at the level of the neuromuscular interface were quantified: stimulus time, defined as the time from the steepest drop in DBP to the first increase in MSNA; and response time, defined as time from the last bust of MSNA to the end of the DBP pulse response. The number of MSNA bursts increased with LBNP level, as would be expected with increased sympathetic drive (331), whereas stimulus time and response time decreased. Furthermore, in the five subjects who were unable to complete the full protocol (non-finishers), the stimulus time was significantly greater compared with those who were able to withstand −60 mmHg LBNP without presyncopal symptoms.

Closely synchronized oscillatory sympathetic control of blood pressure reflects an appropriate compensatory response as a protective mechanism for sustaining adequate oxygenation to the brain and peripheral tissues with progressive LBNP. When this coupling is lost, so is the ability to regulate blood pressure (67, 77)

5. Hormonal responses to LBNP

LBNP has led to a greater understanding of the compensatory hormonal responses to central hypovolemia. These include rapid elevations of plasma epinephrine and norepinephrine and, after a 10- to 20-min delay, renin-angiotensin system activation with elevated plasma renin activity, angiotensin II, and aldosterone (72, 87, 127, 171, 174, 327).

More than 50 yr ago LBNP-induced inhibition of low pressure baroreceptor activity was reported to increase the hypothalamic-pituitary hormone arginine vasopressin (AVP) (318); however, later studies indicated that the simultaneous unloading of low pressure cardiopulmonary and arterial baroreceptors leads to the elevations in AVP (157, 222). Further studies showed that low levels of nonhypotensive LBNP associated with selective unloading of cardiopulmonary mechanoreceptors did not lead to plasma AVP elevations (276). Norsk et al. (276) investigated the effects of changes in pulse pressure during LBNP on vasopressin and renin levels. They observed that LBNP without a decrease in mean arterial pressure (MAP) (which could be associated with inhibition of arterial baroreflexes) leads to release of AVP and renin simultaneously with the occurrence of pulse pressure reductions. This is a particularly important finding with clinical application, as it is possible that the reduced pulse pressure in patients with congestive cardiac failure may be responsible for the elevated vasopressin and renin release (276).

With respect to other hormones, elevated plasma levels of the 52-amino acid peptide adrenomedullin and the 30-amino acid neuropeptide galanin have been identified during LBNP (171, 174). Adrenomedullin has been shown in younger subjects to increase in an intensity-dependent man-

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ner during the first minutes of central hypovolemia (320); such relatively rapid plasma hormone concentration changes are generally indicative of altered release rates into the circulation (171), supported by a long circulating half-life of 22 min (258). In contrast, significant galanin surges have been identified after some delay, possibly due to general spillover from the synaptic clefts of activated neuronal systems, and are rapidly eliminated resulting in a half-life of 4 min (discussed in Ref. 174).

BOX 2 below depicts the temporal sequence of autonomic and hormonal responses which occur from application of low levels of LBNP until development of presyncope [based on Rowell (321), Johnson et al. (193), and Norsk et al (276)].

6. Coagulation and LBNP

States in which increases in bleeding occur (e.g., trauma, rupture of vessels, etc.) have been known to be associated with an increase in thrombosis, which in turn decreases the risk of exsanguination. LBNP-induced central hypovolemia is associated with hemoconcentration and increases in plasma proteins and blood viscosity; all of these create a prothrombotic state. To what extent, however, LBNP induced hypovolemia affects coagulation cascade has only been mostly studied in the last decade (372, 398, 399). Zaar et al. (399) reported that exposing healthy subjects to 10 min of −30 mmHg LBNP leads to activation of the thrombin generating part of the coagulation system as seen in the increases in thrombin anti-thrombin (TAT) complex levels. They suggested that the increases in TAT levels seen during LBNP were similar to those seen in patients with deep venous thrombosis and that the fluid shifts to the legs that occur during the LBNP application (approximate average values of 650 ml) used in their study is similar to those associated with prolonged sitting during bus or aircraft travels. Van Helmond et al. (372) compared the effects of LBNP of up to −45 mmHg with blood loss using blood removal at the rate of 43 ml/s (up to 1000 ml) and observed that slightly greater changes are seen in coagulation markers during LBNP and blood loss. Despite hemoconcentration and hemodilution occurring during LBNP and bleeding, respectively, the coagulation changes were similar in both cases. It was suggested that the increased sympathetic activity observed in both LBNP and bleeding could have contributed to activation of the coagulation via stimulation of the endothelial beta-2 adrenergic receptors (discussed in Ref. 372).

Cvirn et al. (83) have reported that the development of presyncope is also associated with an increase in coagulation. At presyncope, plasma volume losses and the accompanying hemoconcentration changes (which lead to increases in plasma proteins and blood viscosity) are accompanied by elevations in endothelial activation markers (e.g., tissue plasminogen activator and tissue factor) as well as thrombin generation parameters (e.g., prothrombin fragments 1 and 2 and thrombin-antithrombin complexes). Could the increases in vasopressin at presyncope lead to the hypercoagulable state? Vasopressin has been shown to be associated with increases in platelet agglutination and aggregation (108). More recently, Zaar et al. (398) have reported that LBNP-induced presyncope is associated with increased platelet activity and aggregation thus leading to increases in coagulation.

BOX 2. Temporal sequence of autonomic and hormonal responses

- During low levels of LBNP, the cardiopulmonary baroreflexes are inhibited leading to peripheral vasoconstriction and no increase in heart rate
- With increasing levels of LBNP or with longer LBNP exposure, arterial pulse pressure is decreased leading to heart rate increase without demonstrable changes in mean arterial pressure
- When mean arterial pressure falls, then hormone releases (AVP, renin) increase significantly (85, 327), and then finally,
- Syncope ensues (detailed in sect. IIIC)

B. General Principles and Considerations of LBNP Application

Important aspects to consider when using LBNP are reviewed below.

1. LBNP responses depend on the amount of LBNP applied and time of its application (263, 396)

The time and magnitude of hypotension that occurs during an LBNP application vary among subjects (178). It has been estimated that ~0.5–1 liter of fluid is redistributed from central to peripheral vascular compartments when −20 mmHg LBNP is applied for 5 min (78). The amount of blood pooled in the legs during LBNP is also variable across individual subjects.

Different levels of LBNP produce quantitatively different responses, time courses, and dose-response patterns (215). Summarized below is an overview of the effects induced by varying LBNP durations and magnitude.

- LBNP of −10 mmHg reduces central venous pressure (CVP) by 3 mmHg, whereas LBNP of −60 mmHg reduces CVP by 7mmHg: ≈slope of CVP change is 1 mmHg per −10 mmHg LBNP (78);
- Autonomic nervous system responses dominate during short periods (3–5 min onset) of LBNP (241), while longer-duration LBNP exposures (20 min) result in additional neurohumoral responses (175, 263). As such, short-term LBNP application has the experimental advantage of isolating autonomic from humoral responses (286);

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- Graded LBNP causes a reduction in baroreflex responses (64, 66, 68, 75) and has been used to study the integrity of vasomotor reflexes (16).

Thus, when carrying out any study involving LBNP, it is necessary to take into account which LBNP protocol will affect which hemodynamic and neurohormonal responses (see sect. II*B4*).

2. LBNP magnitude affects blood flow in different regions

LBNP, through prolonged baroreceptor unloading, alters regional blood flow (178). For example, LBNP at levels greater than –20 mmHg leads to a different amount of blood pooling in different body regions. Even higher levels of LBNP (greater than –60 mmHg) primarily lead to renal vasoconstriction and reduced renal plasma flow, glomerular filtration rate, and urine production (88, 178, 310, 396).

The largest regional vascular conductance, and an important vascular reservoir, is the splanchnic region (101). Splanchnic vascular reserve appears to help in blood pressure regulation during hypovolemia (22). LBNP up to −20 mmHg decreases splanchnic perfusion but not aortic pressure (178, 193). LBNP of −50 mmHg has been reported to reduce splanchnic blood flow by 32% and splanchnic vascular conductance by 30% (323), primarily due to increased sympathetic activity. In addition, blood flow in the splanchnic region continues to decrease as LBNP increases (101, 193). However, these investigators observed no differences in the decline of splanchnic blood flow when comparing LBNP levels between −20 and −50 mmHg with those below −20 mmHg. Overall, studies on compartmental blood volume distribution during LBNP have revealed highly variable results.

3. Responses to LBNP magnitude have variable effects on different receptors

In humans, cardiopulmonary baroreceptor stimulation results in vasoconstriction in the cutaneous, muscular, and renal regions (384), while arterial baroreceptors mainly influence mesenteric perfusion (178). LBNP-induced stimulation of both high- and low-pressure receptors elicits complex heart rate and vasodilatation reactions in skin, muscle, and splanchnic areas to stabilize blood pressure. Varying the LBNP intensity induces dissimilar types of reflex responses due to variable effects on arterial and cardiac baroreceptors (178, 285, 311). Higher suction, for example, affects both the arterial and cardiopulmonary receptors (17, 120), thereby reducing forearm and splanchnic blood flow (178) and pulse pressure (89). In addition, Bevegård et al. (20) suggested that intrathoracic receptors sense blood volume distribution changes and modify carotid baroreflex function.

4. Constant LBNP application versus increasing LBNP leads to different effects

TABLE 2 illustrates various LBNP protocols and the physiological effects of varying LBNP levels and durations (138). As the LBNP-induced effects depend on LBNP levels and duration, Goswami et al. (138) proposed categorization of protocols based on LBNP level, duration, and type of application (constant LBNP vs. increasing LBNP).

5. LBNP sealing location affects physiological responses

LBNP sealing location determines the extent of decrease in central blood volume in response to LBNP (136). The default position at which the LBNP seal is applied is at the iliac crest. LBNP application at this position does not cause mechanical compression of the urinary bladder or intestinal region and, consequently, does not compromise splanchnic blood flow (31).

Goswami et al. (136) investigated further the effects of two varying sealing positions: upper abdomen (UA) and iliac crest (IC) during application of −40 mmHg LBNP. They observed that the position of the seal used for LBNP affected cardiovascular responses. Some of these differences were observed even before the application of the LBNP, suggesting mechanical effects of seal location. In addition, applying LBNP with the seal located at upper abdomen led to an elevation in heart rate and significantly greater decreases in central blood volume, splanchnic blood flow, and greater sequestration of blood into lower body regions (136). The greater shift of the blood into the lower body regions is supported by the findings that concomitant femoral vein distension and caval vein narrowing occurs during compression of the abdomen (343).

III. LBNP TOLERANCE

One of the common outcomes of central hypovolemia induced by progressively increasing levels of LBNP is the onset of presyncope that is usually identified by a precipitous reduction in MAP below −60 mmHg concurrent with expressions of symptoms such as tunnel vision, sweating, nausea, or dizziness (60, 331). The LBNP tolerance of a human subject is defined by the total time, or a product of the total time and LBNP level (cumulative stress index, CSI), required to reach a presyncopal end point (for details, see Refs. 55, 60, 331). As such, LBNP tolerance is assessed by occurrence of a presyncopal end point and then quantifying LBNP level with the CSI (240), LBNP tolerance index (LTI) (182), or final LBNP pressure attained (333).

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Table 2. *LBNP protocols and effects of different LBNP levels*

Pressure Protocol	Variables Studied	Specific Effects
Mild level, constant LBNP pressure, short duration	1. Unloading of cardiopulmonary receptors 2. Mild cardiovascular stress 3. Mild hemorrhage	**−10 mmHg LBNP** Unloads cardiopulmonary baroreceptors: CVP decreases by 3 mmHg, no effect on MAP, pulse pressure, or heart rate Peripheral vasoconstriction in upper, lower limbs. Minimal arterial baroreceptors effects. **−20 mmHg LBNP for 5 min** 500–1,000 ml fluid displacement from central to peripheral vascular compartments. Increases in leg volume by 450 ml.
Mild level, constant pressure, long duration	1. Leg interstitial fluid pressure changes, plasma volume, and net whole body transcapillary fluid transport affected 2. Splanchnic blood flow	**10 mmHg LBNP for 60 min** Forearm, splanchnic blood flow reduced; renal blood flow unchanged. **30 mmHg LBNP for 4 h** Affects leg interstitial fluid pressure.
Mild level, pressure increase (ramp), short duration	1. Skin, muscle sympathetic nerve activity, and blood flow in forearm, leg 2. Selected hormonal responses	Progressive increases in sympathetic activity, in upper, lower extremities comparable to cold pressor test and static exercise. Near-maximal vasoconstrictor responses in limbs at 3 min. Short periods limit humoral responses; increases in plasma norepinephrine at 3 min.
Mild level, pressure increase (ramp), long duration Moderate/strong level, constant LBNP pressure, short duration	Responses of renin and aldosterone 1. Hypovolemia and effects of integrated cardiopulmonary and arterial baroreflex response to moderate cardiovascular stress 2. Splanchnic blood flow 3. Differential blood filling in body compartments 4. Selected hormone releases 5. Cerebral blood flow affected 6. Renal flow and urine output 7. Plasma volume expansion 8. Differences between the sexes, as 50 mmHg is different relative LBNP stress for each sex	PRA, aldosterone increase after 20 min of −10, –20 mmHg LBNP **Greater than –20 mmHg LBNP** Differential blood filling in different body compartments. **−40 mmHg LBNP for 5 min** 614 ml fluid displacement from central to peripheral vascular compartments. Leg volume doubles when LBNP increases from −20 to −40 mmHg. Increase in systemic vascular resistance. Carotid pulse pressure decreases. Similar responses to upright posture. Not sufficient for release of AVP. **−40 mmHg LBNP for 10 min** Cerebral blood flow reductions but no change in middle cerebral artery diameter. **−50 mmHg LBNP for 5 min** Sex differences; women showing lower tolerance. 32% fall in splanchnic blood flow. 30% reduction in splanchnic vascular conductance. Return of forearm flow to resting levels. **−60 mmHg LBNP** CVP falls by 7 mmHg. Forearm muscle sympathetic activity doubles. Changes in peripheral venous tone. Further decrease in splanchnic blood flow. Decreases in renal plasma flow, GFR, and urine output **−80 to −90 mmHg for 5 min** Rapid decrease of central blood volume. Plasma volume expansion (700 ml in 5 min) by fluid transfer from muscles and skin to circulation.
Moderate level, constant LBNP pressure, long duration	1. LBNP tolerance 2. Presyncopal studies: subjects become presyncopal at different LBNP levels	**−55 mmHg LBNP for >20 min** Increase vagal activity during LBNP induced hypotension.

Continued

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Table 2.—*Continued*

Pressure Protocol	Variables Studied	Specific Effects
Moderate/strong level, pressure increase (ramp), short duration	1. Hormonal patterns 2. Cardiovascular responses to moderate stress	Near-maximal vasoconstrictor response. Higher magnitude of norepinephrine at 3 min.
Strong level, pressure increase (ramp), long duration	1. Cardiovascular responses to severe stress 2. Study sympathetic skeletal muscle and skin resistance vessel control 3. Blood pressure responses to carotid baroreceptor stimulation	Stimulates more significant volume contraction and unloads cardiopulmonary and arterial baroreceptors with decreases in splanchnic and renal blood flow. Activates renin-aldosterone system, increases filtration fraction and forearm blood flow. Enhanced sympathetically mediated blood pressure responses to carotid baroreceptor stimulation.

LBNP, lower body negative pressure; GFR, glomerular filtration rate; CVP, central venous pressure; PRA, plasma renin activity; AVP, arginine vasopressin. [Modified from Goswami et al. (138).]

A. Typical Responses to LBNP Up to Presyncope

FIGURE 4 demonstrates a characteristic response of MAP and heart rate from a healthy male subject undergoing a graded LBNP protocol up to presyncope (55). In this protocol, LBNP was initiated at −20 mmHg for 3 min and gradually increased by −10 mmHg every 3 min. Up to −30 mmHg LBNP, blood pressure, and heart rate remained stable. At −40 mmHg LBNP, blood pressure reductions occur but are corrected by compensatory reflex increases in heart rate and systemic peripheral vascular resistance. Greater increases in LBNP lead to the development of hypotension, which continues until compensatory mechanisms cannot sustain arterial blood pressure and subjects develop presyncope (55). Presyncope is accompanied by a vasovagal reaction characterized by a dramatic and rapid development of hypotension and bradycardia. The initial reductions in arterial blood pressure occur before the fall off in heart rate (discussed in Ref. 55), but the final rapid onset of hemodynamic failure coincident with pronounced hypotension is preceded by a bradycardia (331).

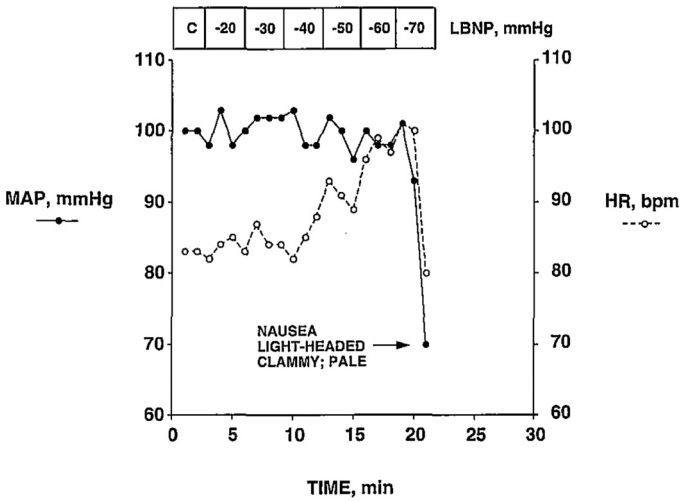


FIGURE 4. Responses of mean arterial pressure (MAP, closed circles and solid line) and heart rate (HR, open circles and dashed line) during graded presyncopal lower body negative pressure (LBNP) in a young healthy subject. [From Convertino (55).]

B. Autonomic Responses at Presyncope During LBNP

An "empty heart" that results from low ventricular filling has been proposed as an underlying mechanism for a so-called "vasovagal" syncope (339). There is evidence that supports the notion that a resulting cardiac receptor stimulation initiates a Bezold-Jarish reflex-mediated bradycardia via afferent vagal activation (381a). Bradycardia and vasodilation at the time of LBNP-induced syncope have also been associated with sympathetic nerve activity withdrawal that appears to be one contributing physiological mechanism to hemodynamic instability in humans (34, 49, 74, 77, 188, 189, 199, 267, 328, 331, 374, 380). In addition to sympathetic withdrawal, syncope during LBNP is also associated with the onset of disruptions in normal autonomic function that includes failure to sustain coherence between arterial pressure oscillations and sympathetic nerve activity and sympathetic nerve firing characteristics that manifest as fused integrated bursts (67, 77). As such, it has been proposed that LBNP-induced hypotension and syncope are preceded by reduced baroreflex control over sympathetic neural activity (184).

C. Hormonal Changes Associated With Presyncope/Syncope

Presyncope is associated with increased vasopressin secretion (15, 85, 127, 174, 327), stimulating subsequent secretion of adrenocorticotropic hormone and cortisol (171). The rise in AVP seems to be greater in less tolerant people undergoing levels of LBNP similar to those of individuals

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with high tolerance (144), but higher tolerance individuals display greater AVP levels at presyncope (57, 72, 331).

Other hormonal responses seen at presyncope include a threefold elevation in plasma renin activity (PRA) above supine control (85, 123). PRA was found to be higher in persons who tolerated levels of LBNP at −50 mmHg (18 min) well (144) or even greater levels (72). A reduction in atrial natriuretic peptide (ANP) at presyncope has been observed (123). Furthermore, increased plasma galanin and adrenomedullin levels have been seen at presyncope. There appears to be an association between the elevated plasma galanin levels and plasma adrenomedullin levels as well as the accompanying vasodilation and occurrence of hypotension (171). However, this needs to be explored further.

D. LBNP Tolerance: Repeatability and Assessment

Reproducibility of the physiological responses induced by perturbations is particularly important when deciding which device or method is to be used for any physiological study. LBNP-induced physiological responses, as well as LBNP tolerance, have been shown to be highly reproducible if the same protocol is conducted under similar physiological states with testing carried out at the same time of the day (55, 182, 230). That is, despite LBNP tolerance showing wide variance between individuals, there is a high intra-individual reproducibility of LBNP tolerance. For example, application of graded LBNP to the same individuals four times (with each run separated by at least 2 wk) did not lead to changes in tolerance times across trials (137, 230).

During repeated runs of graded LBNP, consistent patterns in heart rate and stroke volume are seen, suggesting that stimulus-response relationships are characteristic for a given individual. On the other hand, patterns that clearly deviate between persons suggest different individual-specific strategies for maintenance of blood pressure during central hypovolemia. Indeed, the response gains for the various actuators within baroregulatory pathways (e.g., heart rate, heart contractility, peripheral resistance, and venous compliance) appear to be different while acting to reach the same goal of maintaining blood pressure for adequate oxygen delivery to tissues (196). For instance, the rate of change (e.g., increases or decreases) in heart rate or stroke volume is different between individuals (137). It appears that some individuals use different strategies to stabilize arterial pressure in different experimental runs (38). This would seem to confirm earlier observations that the cardiovascular regulatory system "searches" for optimal responses to maintain appropriate blood pressure and, hence, perfusion levels (196). However, long-term stability in regulatory behavior does not preclude short-term training effects, as observed in pilots repeatedly exposed to high-G flight profiles which results in improvements in their G-tolerance, even though for only short periods of time (143).

Hemodynamic, neuroendocrine, and tolerance responses are highly reproducible even over long periods (**FIGURE 5, *A–C***, respectively; Ref. 55). Data obtained from seven healthy males (age 28–49 yr) who repeated LBNP test protocols 1 yr apart (detailed in Ref. 55) under similar physiological conditions (e.g., time of day, hydration status, physical activity, meal composition, and fluid intake) and experimental control of other factors that could potentially affect LBNP-induced responses suggest this. These data were collected before the sudden onset of vasovagal responses (e.g., bradycardia) and hypotension with MAP falling below 80 mmHg despite elevations in heart rate (>100 beats/min) and peripheral vascular resistance (>80 units) (**FIGURE 5*A***) (55). There was also a reproducibility in the response of plasma norepinephrine, vasopressin, and renin-angiotensin to LBNP-induced presyncope from year 1 to year 2 (**FIGURE 5*B***). As such, LBNP can be used to predict subjects who are at higher risk of fainting or early onset of hemorrhagic shock (55). This is detailed further in section VC2.

FIGURE 5*C* shows that the difference between high and low tolerant individuals is also repeatable (55). On the other hand, some researchers also reported that individuals become adapted to repeated LBNP exposure, with LBNP tolerance increasing by the third successive exposure (182). As some studies did not clearly define the end point of LBNP tolerance, differences in results could be due to the ambiguity associated with determination of the LBNP tolerance end point (182) or not taking into account the lay-off effect when testing is conducted more than 1 wk apart. It is known, for example, that pilots who have not flown at high G for over a week develop decreased G tolerance upon return to flying (143), a condition known as "G-layoff".

E. Predictors of LBNP Tolerance

Some factors that have been identified as possible "predictors" of LBNP tolerance include heart rate and cardiac reserve (329), alterations in fluid homeostasis and volume regulating hormones (144), blood volume (22), inotropic and autonomic factors (397), and the intensity of splanchnic arterial vasoconstriction (8). There is evidence that less intense splanchnic arterial vasoconstriction is associated with a higher occurrence of syncope (8). Arbeille and Herault (6) used Doppler ultrasound measurements to quantify the cerebral to femoral flow redistribution during central hypovolemia induced by LBNP. They measured blood flow, flow volume, and vascular resistance in the cerebral and femoral arteries. Their studies show that the femoral resistance index (reverse/forward velocity) and the cerebral-to-femoral flow ratio (cerebral/femoral mean velocity) are accurate predictors for LBNP intolerance (6).

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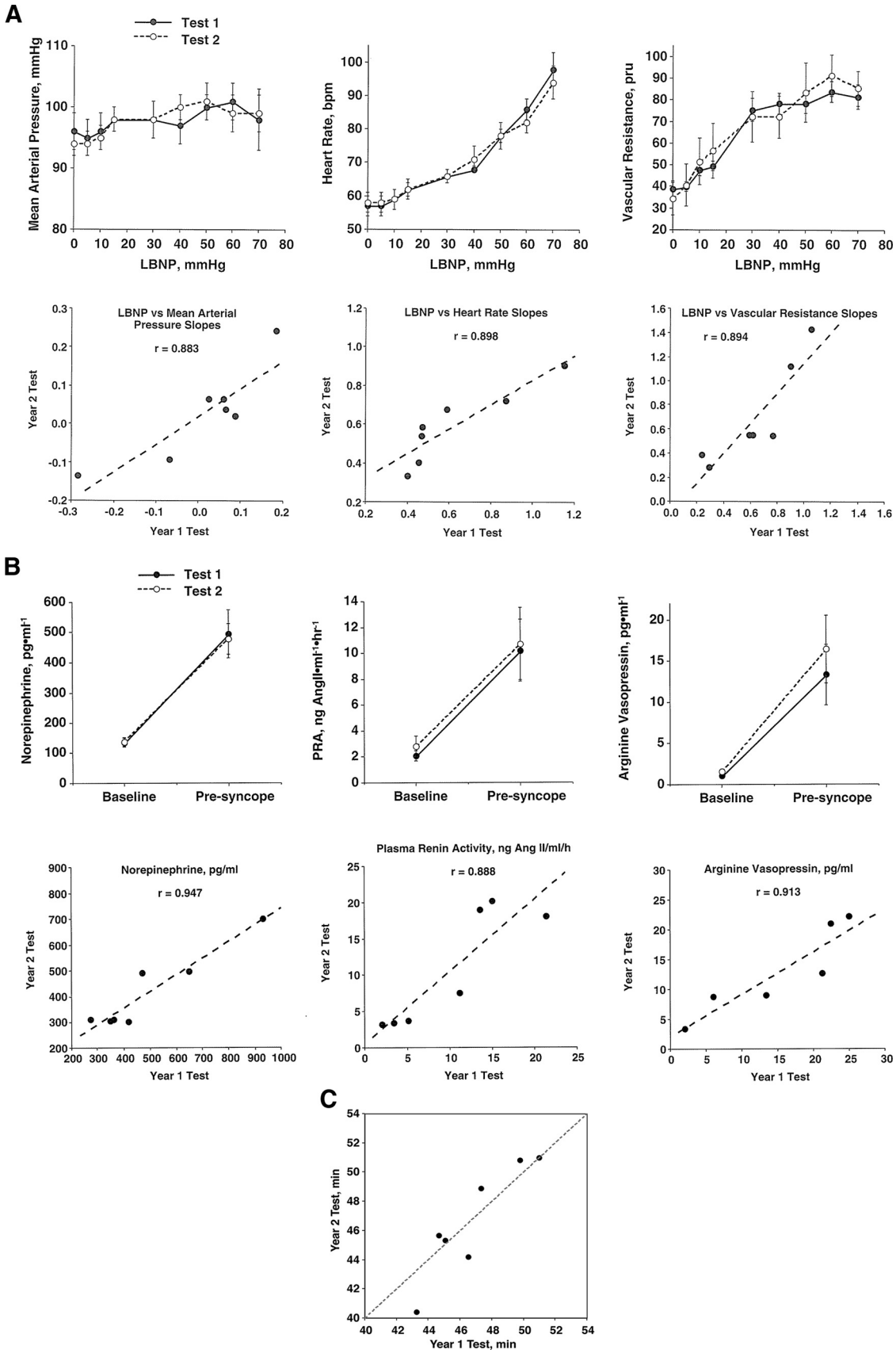


FIGURE 5. *A–C*: reproducibility of mean arterial pressure, heart rate, and vascular resistance (*A*), selected hormones (*B*), and orthostatic tolerance (*C*) during repeated lower body negative pressure (LBNP) testing separated by 1 yr. [From Convertino (55).]

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The usage of heart rate or heart rate reserve as predictors of LBNP tolerance, however, is not well established as some individuals respond with exaggerated elevations in heart rate (have a presyncopal heart rate near maximum heart rate), while others are non-heart rate responders (have presyncopal heart rates of only ~130) (383).

Despite decades of experiments designed to identify factors associated with tolerance to central hypovolemia, attempts to incorporate single or multiple physiological responses into multivariate models designed to "predict" LBNP tolerance have not proven successful. The failure to accurately predict LBNP tolerance in individuals has been the result of the use of population averages that mask individual variability of compensatory responses to central hypovolemia. This concept was eloquently demonstrated in a study by Carter et al. (38) who found that LBNP tolerance was accurately predicted by an algorithm that had the capability to assess the total integration of compensatory mechanisms independent of significant individual subject variability in single compensatory responses (e.g., heart rate, stroke volume, peripheral vascular resistance, sympathetic nerve activity). In other words, accurate prediction of LBNP tolerance requires the capability to measure the unique physiological strategy of total integration of all compensatory mechanisms in each individual. The use of LBNP has allowed for the development of such a capability.

F. LBNP Tolerance and Its Application in Investigational Research for Comparison of Subject Populations

Due to its high intraindividual reproducibility, LBNP tolerance testing can be applied for studying underlying mechanisms in varying patient conditions or the same pre- and post-interventions (e.g., bedrest, spaceflight, etc.). LBNP has been used for classification of healthy individuals into high tolerance and low tolerance groups as well as for identification of predictors for low tolerance. Echocardiographic measurements of cardiac function during LBNP showed that both high and low tolerance groups had comparable presyncopal stroke volume, ejection fraction, and end-diastolic and end-systolic volumes (i.e., cardiac filling and emptying).

High LBNP tolerance individuals undergoing LBNP exhibit the following:

1. Greater capability to increase heart rate (subsequently able to maintain cardiac output) and systemic peripheral resistance (72, 329);
2. A larger increase in vascular resistance, despite peripheral vascular resistance being similar at rest (329);
3. Higher circulating plasma concentrations of vasoactive hormones (norepinephrine, vasopressin, and renin-angiotensin) at the time of presyncope accompanied with a greater vasoconstrictive reserve (57, 72, 144, 331);
4. Greater arterial baroreflex responses and circulating blood volume (238).

Interestingly, studies investigating blood pooling in the lower legs have found that subjects with high tolerance to LBNP have equal or greater accumulation of volume when compared with subjects with low tolerance at similar LBNP pressures (153, 329). These data do not lend support to the assumption that individuals with low LBNP tolerance have increased venous pooling in the legs compared with individuals with high LBNP tolerance. However, increased pooling in other regions, such as the splanchnic region is a distinct possibility. Hachiya et al. (153) also reported a delayed onset of oxygenated Hb reductions (as measured by near-infrared spectroscopy) in response to blood pooling in individuals with high tolerance to central hypovolemia. In this study, oxygenated Hb was used as an index of arterial vasoconstriction in the calf (151). A follow-up study (22) provided further evidence that this delayed response was presumably due to the mobilization of reserve blood volume, mainly from the splanchnic region (8), or greater vasoconstrictor capacity (74).

Evans et al. (104) found a negative relationship between hematocrit change and LBNP tolerance. The pattern of hematocrit changes during and following LBNP suggested that the difference between syncopal and nonsyncopal response was due to differences in venous pooling as opposed to differences in plasma filtration rate. This would indicate a difference in compartment storage in the two groups exhibiting different LBNP tolerance times; however, new studies (8, 22, 152, 153) suggest that the observations of Evans et al. (104) could be attributed to a greater ability of people with high tolerance to sequester red blood cells in the lower limbs without compromising central blood volume.

The response mechanisms that contribute to LBNP tolerance can, however, remain hidden by use of standard clinical observational practice. This was clearly demonstrated in a series of studies by Hachiya et al. (152) who noted that vasoconstrictor and heart rate responses to LBNP were delayed in participants with high LBNP tolerance compared with those with low LBNP tolerance. Such differences can be attributed to a greater ability for internal blood volume redistribution to central circulation in participants with greater tolerance to central hypovolemia (22), suggesting a greater blood volume reserve that could act to delay their compensatory response.

G. Factors Affecting LBNP-Induced Responses and LBNP Tolerance

TABLE 3 outlines some confounding variables that could potentially affect LBNP-induced responses and tolerance.

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Table 3. *Effects of confounding variables such as subject selection criteria and LBNP protocols on physiological responses during LBNP*

Variable	Possible Effect	Physiological Basis
	Subject selection criteria	
Sex	Women have lower tolerance to various orthostatic challenges	Possible etiology: heart rate responses, vagal withdrawal, differences in total peripheral resistance, cardiac filling, cardio-vagal baroreflex, baroreflex latency period, regulation of cerebral perfusion, hormones.
Menstrual cycle and orthostatic intolerance	Menstrual cycle affects orthostatic intolerance	Contradictory data reported.
Age	Aging is associated with orthostatic hypotension and falls (131, 134)	Orthostatic challenge in older persons is accompanied by limited heart rate and cardiac output response, reduced baroreceptor efficiency, stroke volume variability, venous compliance; all these contribute to lower tolerance. Furthermore, aging modulates central sympathetic control of the arterial, cardiopulmonary baroreflex, and vestibulo-sympathetic reflex.
Height	Height could affect orthostatic tolerance	An inverse relationship between height and LBNP tolerance has been reported. However, a recent study found no relationship between anthropometry and G-tolerance in fighter pilots (289).
Fitness levels	Fitness is related to orthostatic intolerance Different types of training affect orthostatic challenge differently	Fitness training can lead to lower orthostatic tolerance, improvement in orthostatic tolerance, or can have no effect. Exercise training causes increased limb compliance, ventricular hypertrophy, increases in total blood volume, and reduced carotid and aortic baroreflex gains thus leading to reduced tolerance. Highly trained persons have alterations in their ventricular response. Orthostatic intolerance is greater in athletes that are trained upright compared with those who are trained supine.
	Variation in experimental procedures and protocols	
Time of day	Circadian variations affect LBNP-induced responses (129, 287)	Lower forearm vascular conductance and stroke volume in the morning. Higher blood pressure in the morning, due partly to the increased alpha sympathetic vasoconstrictor activity.
Prandial state	Consumption of food affects LBNP-induced responses (84, 359)	Ingestion of food increases heart rate and cardiac output, primarily mediated by neurally mediated sympathetic nerve activity. Varying meal composition (carbohydrates, fats, and proteins) alters splanchnic or portal vein blood flow or the time delay of the hemodynamic response. LBNP-induced sympathetic nervous excitation has varying metabolic effects on glucose and lipid oxidation, during postprandial state.
Fluid and sodium intake	Issue of body sodium stores vs. blood volume and cardiovascular regulation remains unresolved (163, 342)	Dehydration increases hemodynamic and plasma renin activity responses to central hypovolemia, especially in low tolerant subjects (162). Aldosterone is also elevated in dehydrated humans (348).
Pre-LBNP supine baseline period	Sufficient time should be given to ensure a resting baseline level	Resting levels of sympathetic activity must be stabilized, as it influences reflex-mediated changes in end organs (e.g., muscle nerve activity).

Continued

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Table 3.—*Continued*

Variable	Possible Effect	Physiological Basis
Body position	Postural changes during LBNP affect cardiovascular responses	Prone position is associated with a larger stroke volume reduction and the lateral position with a greater decrease in mean arterial pressure (342). Renin activity and aldosterone levels also change with posture. Stricter postural control must be maintained to ensure measurement consistency (342).
Sealing location	Iliac crest is the "default" LBNP seal location	Seal around the iliac crest does not mechanically compress the bladder or affect splanchnic blood flow. Mesenteric/hepatic blood flow before and during LBNP may be affected by increased intra-abdominal pressure. As splanchnic region may be involved in blood pressure regulation during central hypovolemia induced by LBNP (25, 136, 173), it is important to ensure that the LBNP sealing is at the iliac crest.

[Summarized from Goswami et al. (138).]

These should be considered when designing physiological experiments involving LBNP. Some of these aspects are discussed here in detail.

1. Sex

As more women are being selected for military and space missions, understanding sex differences that exist in physiological mechanisms that underlie blood pressure regulation is becoming more important. There is compelling evidence that the general population of women has a lower tolerance to central hypovolemic challenges when compared with men (37, 54, 135, 176, 387, 388). Underlying physiological mechanisms include lower stroke volume, a larger rate of decrease in cardiac output and MAP, and greater proportional use of reserve capacity to elevate peripheral vascular resistance (i.e., to elicit maximal arterial vasoconstriction) in females compared with males. Hachiya et al. (152) reported that females have reduced vasoconstrictor response (i.e., reserve) to a given accumulation of calf blood volume during LBNP application compared with males. Furthermore, the lower tolerance time to central hypovolemia in females as compared with males could be attributed to differences in adrenergic responses during higher LBNP levels (54, 204, 265, 387). While this has also been observed in females post-spaceflight (117, 118), the mechanism responsible for these differences in adrenergic responses is currently not clear (113, 114, 117).

Some authors have reported that heart rate responses during similar central hypovolemia application in men differ from responses in women (204). Women may primarily respond with vagal withdrawal (116), while men show a greater sympathetic response (107, 116) and a larger increase in total peripheral vascular resistance (115). Additionally, cardiac filling, cardiovagal baroreflex, and baroreflex latency period have also been shown to be different in males and females (19, 119, 146). Women have also shown greater pelvic blood pooling than men (156, 388). However, some studies did not report any difference in cardiovascular responses and tolerance to low (304) or even maximal LBNP loading (165, 204) across sex.

A factor that could also play an important role in sex differences associated with presyncopal responses is cerebral perfusion regulation. Since females normally exhibit higher levels of ventilation, they typically have reduced CO_2 blood levels compared with males (234). It is plausible, therefore, that hyperventilation during central hypovolemia will further decrease blood CO_2 levels and thereby reduce cerebral perfusion in females as compared with males. Indeed, inspiring 5% CO_2 has been shown to increase the time taken to reach presyncope across both sexes (181).

A study of 27 shuttle astronaut responses to standing has provided evidence that there exist possible sex differences in cerebrovascular regulation (25). Zuj et al. (405) found that dynamic cerebrovascular CO_2 reactivity during supine rest was impaired in the seven International Space Station astronauts. In this study astronauts were also exposed to mild LBNP (max −20 mmHg) pre- and post-flight and their cerebrovascular responses were found to be normal. To explore whether these changes in dynamic cerebrovascular reactivity have pathophysiological significance, especially in relation to complications in vision arising during spaceflight induced by cephalad fluid shifts (401), further investigation at higher levels of LBNP is required pre-flight, in-flight, and post-flight.

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A) INFLUENCE OF SEX ON HORMONAL RESPONSES. LBNP elicits similar elevations of vasopressin and renin-angiotensin in males and females (54). Sex has been associated with lower levels of catecholamines (54, 127, 386), adrenomedullin (202), and galanin (174) at rest and during LBNP. Lower plasma adrenomedullin levels in females supports the notion that sex steroids may influence a regulatory role in adrenomedullin production by human endothelial cells (253). Similarly, a larger presyncopal increase of plasma norepinephrine was observed in males (54), while greater increases in galanin were seen in females at presyncope (174). Galanin, a neuropeptide, has been shown to play an important role in cardiovascular regulation and the general stress response.

In addition, differential hormonal responses to central hypovolemia across the sexes presumably relate to a modulatory effect of sex hormones (294), with an assessment of such differences complicated by hormonal changes during the normal female menstrual cycle (261). Data related to the effects of the menstrual cycle on the time it takes to reach presyncope, however, are contradictory (45, 204). It remains to be established if the phase of the menstrual cycle influences presyncopal responses and LBNP tolerance.

While sex steroids could play a pivotal role in cardiovascular and hormonal responses during graded central hypovolemia in young females, the extent to which menopause-associated decrease in plasma estradiol affects neurohormonal responses to central hypovolemia induced by standing up or LBNP is currently not known. As falls and fall-related injuries are more common in older females in comparison with age-matched males (352), there is a need to further understand the mechanisms that contribute to falls and fall-related injuries in older females, especially during central hypovolemia which can be studied using LBNP.

2. Age

Aging is associated with orthostatic hypotension (131, 134, 324), lower heart rate and cardiac output response to central hypovolemia (111, 165), reduced baroreceptor sensitivity (281) and venous compliance (279, 280), lower stroke volume variability (147), and reduced time to reach presyncope during central hypovolemia (231). In addition, resting supine blood pressure, heart rate, and norepinephrine levels appear to be influenced by age (373). However, both younger and older males appear to respond to reductions in central venous pressure during progressive LBNP with similar reductions in stroke volume and increases in heart rate, total peripheral resistance, and forearm vascular resistance (373).

As age increases, the central sympathetic control of the arterial (145, 251, 335) and cardiopulmonary baroreflexes (46) becomes decreased in both normotensive (272) and hypertensive humans (211). On the other hand, some authors have reported an augmentation of these reflexes with age in healthy humans (86). Davy et al. (86) observed lesser reflex-mediated increases in limb vascular resistance during LBNP-induced central hypovolemia in older adults due to the attenuated vasoconstrictor response to sympathetic stimulation. Furthermore, the gain (sensitivity) of the vestibulo-sympathetic reflex is reduced with age (308) as well as during central hypovolemia induced by LBNP (264). The extent to which these age-related changes in vestibular function affect blood pressure regulation during LBNP needs to be further explored.

3. Height

Physical factors (e.g., height) appear to influence time to syncope during LBNP. Height and LBNP tolerance have been shown to have an inverse relationship (238, 292) which may act to provide a protection against pooling of blood in individuals with a shorter hydrostatic column (239). Ludwig et al. (239) have singled out height as a "most important correlate of $+G_z$, LBNP, and general orthostatic tolerance."

4. Fitness level

Greater limb compliance, ventricular hypertrophy, alteration in ventricular response, greater blood volume, and lower carotid and aortic baroreflex gains have been proposed as some of the factors that contribute to the cross-sectional observations that endurance-trained atheletes with high maximal aerobic capacities (i.e., $\dot{V}O_{2max}$ >60 $ml \cdot min^{-1} \cdot kg^{-1}$) display lower tolerance to central hypovolemia compared with more sedentary individuals (53, 73, 204, 306, 307, 346). However, subsequent LBNP experiments have demonstrated that endurance-trained athletes have greater effective left ventricular compliance and distensibility than non-athletes and a steeper slope of the Frank-Starling relationship between left ventricular filling pressure and stroke volume (224). Consequently, greater reduction in stroke volume for the same reduction in filling pressure represents the most likely mechanism predisposing the endurance athlete to syncope during progressive central hypovolemia induced by LBNP.

In longitudinal studies, regular exercise performed by non-endurance-trained individuals has consistently resulted in either enhanced (53, 286) or no decrement (229) in tolerance to progressive central hypovolemia induced by LBNP. Taken together, these observations support the emergence of a concept that the association of LBNP tolerance with physical fitness is best described as a U- or J-shaped stimulus-response relationship.

5. Seasonal variations

As outlined in the previous sections, vasopressin is a potent vasoconstrictor released during LBNP exposure that is

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known to increase the inhibitory effects of both cardiopulmonary and arterial baroreceptors, and associated with a reduction in sympathetic outflow (207). Seasonal variations have been reported in vasopressin, melatonin, and plasma osmolality (9). In addition, the suprachiasmatic nucleus reduces in size and contains fewer vasopressin-producing neurons in young persons during summer months (180). The extent to which seasonal variations in vasopressin influence LBNP tolerance has, however, not been systematically studied. Further studies are required to investigate the effects of seasons and vasopressin levels during LBNP-induced central hypovolemia on cardiovascular responses and LBNP tolerance, especially across different age groups and sex.

6. Temperature

Temperature changes have been shown to affect physiological responses during central hypovolemia induced by LBNP, with heat stress reducing and cold stress enhancing LBNP tolerance (79). The impact of ambient temperature on venous compliance (396) is most likely a contributing factor to body temperature influences on LBNP tolerance. When ambient temperature rises >28°C, forearm venous compliance has been shown to increase (366, 367) with peripheral vasodilation. The resulting reductions in cerebral perfusion lead to lower LBNP tolerance (395). However, there is considerable variability in LBNP tolerance between individuals exposed to heat (219).

The role skin plays in local heating has been well-studied (40). Even minimal heat application in resting humans activates the cutaneous vasodilator system (340, 396). The decrease in cutaneous blood flow in individuals during LBNP while subjected to heat stress occurs due to neurally mediated mechanisms (340). On the other hand, cooling of the skin causes improvements in LBNP tolerance, as it leads to augmentation of the sympathetic activity (92), peripheral vasoconstriction, and elevated central venous pressure (82). It appears that the temperature of the skin affects both the direct local control and reflex vasoactive state of the skin (396).

Crandall and colleagues (393) carried out an elegant LBNP study in which the interaction between temperature and the Frank-Starling relationship was assessed. They observed direct thermal effects on the control and regulation of stroke volume: heating caused a downward shift of the operating point to a steeper portion of a Frank-Starling curve while cooling led to an upward shift to a flatter portion of the curve. This shift resulted in a larger reduction in stroke volume per reduction in left ventricular end-diastolic pressure (393). Other factors that are modulated by heat stress, and which affect the Frank-Starling relation in persons undergoing LBNP, are heart rate, preload, afterload, and compliance (393). Thermal stress leads to an increase in heart rate via effects on the sinoatrial node and accompanying cardiac sympathetic neural activity but also leads to a reduction in preload and afterload. Combinations of increased heart rate and stroke volume lead to an increased cardiac output during heat stress (393). Whole body thermal stress during LBNP generally leads to reductions in cerebral vascular conductance and pulmonary capillary wedge pressure. On the other hand, cold stress has no effect on heart rate (perhaps due to baroreflex suppression) but leads to elevations in preload and afterload, and consequently, increased LBNP tolerance (393). Moreover, the operating point shifts upward on the plateau portion of a Frank-Starling curve during cooling compared with normothermia (393). In comparison to heat stress, cooling elevates cerebral vascular conductance and pulmonary capillary wedge pressure. The effects on cardiac performance of whole body heating and cooling are presented in a comprehensive review by Wilson and Crandall (394).

7. Sealing skirts and size of waist opening

Various methods (e.g., neoprene skirts) have been used to ensure airtight sealing at the iliac crests. The size of the waist opening is important, as the level of LBNP pressure that produces specific cardiovascular and footward forces depends on waist seal design. For example, Watenpaugh et al. (383) have shown that −100 mmHg LBNP produces a 1.0-G_Z footward force, but with the opening of the LBNP chamber only slightly larger than the subject's waist, the cardiovascular responses are similar to 2-G. However, with an opening about twice the waist circumference, −50 mmHg LBNP produces one body weight footward force and cardiovascular responses are similar to 1-G. This is a crucial advantage of LBNP, as cardiovascular responses and footward forces can be individually manipulated, depending on the aim of the study, to produce a high level of impact with a low cardiovascular strain or vice versa.

8. Saddle/footplate

A saddle provides perineal support and prevents caudal displacement of the subjects. The use of a saddle during LBNP in space flight has even been shown to improve tolerance to central hypovolemia (42). The effect of a footplate is similar because as pressure increases, the leg muscles contract to resist the footward force and can actually raise blood pressure when LBNP pressures exceed −80 mmHg (383).

IV. LBNP AS A TOOL FOR STUDYING CENTRAL HYPOVOLEMIA DURING ORTHOSTATIC CHALLENGE

When standing, ~700 ml of blood can pool in the legs, with half of the total shift occurring within 20–30 s (271, 364). Postural changes to an upright position cause reductions in central venous pressure as well as arterial baroreceptor

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pressures and lead to increases in catecholamines, renin-angiotensin-aldosterone, and sometimes vasopressin (273, 275, 321, 322).

While cardiovascular reflexes associated with upright posture (orthostatic reflexes) can be studied directly with stand tests, most research utilizes passive HUT to minimize interference from muscular reflexes (179, 214, 364). Matalon and Farhi (255) discussed cardiopulmonary adjustments during passive tilting almost four decades ago. HUT of up to 70° has been reported to be associated with increases in heart rate of up to 40%, stroke volume decreases of up to 45%, and up to 27% decreases in cardiac output (233, 368). During HUT, CBF decreases by 10–20%, splanchnic flow by ~40%, and forearm flow by up to 60% (345, 404). There appears to be a correlation between tilt angle and decreases in thoracic blood volume, central venous pressure, atrial diameters, as well as between angle of tilt and magnitude of cardiovascular and hormonal responses (34, 155, 214). For instance, Laszlo et al. (214) have reported that the release of catecholamines depends on the degree of tilt: moderate HUT triggers cardiopulmonary baroreceptor unloading, which elicits reflex sympathoexcitation, while higher degrees of HUT cause greater sympathetic activation and more norepinephrine release. Additional hormonal responses to HUT vary significantly: some authors have reported that renin is released only 20 min following the start of HUT (197) while others have observed increases in plasma renin activity even with low-level HUT (214). The differences in hormonal responses seen across these studies could be due to differences in the tilt angles used.

Although not an orthostatic challenge, LBNP can be used to investigate similar fluid shifts and related hemodynamic responses as those induced by HUT due to the translocation of fluid volume to the lower part of the body. LBNP has the advantage that it, unlike HUT, does not lead to otolith G_z stimulation. HUT is associated with vestibular G_z system (head-to-foot gravitational loading) stimulation with minimal G_x (front-to-back gravitational loading) and angular accelerations (384) and induces variable postural muscle contractions (382). As such, LBNP allows for the study of the isolated contributions of central hypovolemia induced by orthostatic challenges.

A. Similarities and Differences in Physiological Responses to LBNP and HUT

Passive 70° HUT induces a 500- to 600-ml shift of blood to the legs, movement of fluid from intravascular to interstitial space, and decreases in cardiac output, which are similar to those, induced by −40 mmHg LBNP (10, 170, 270, 344, 396). However, as LBNP and HUT are different physiological stressors, they also have different physiological effects (396) which are summarized in **TABLE 4** and discussed in the following sections.

1) Despite both HUT and LBNP being associated with decreases in central blood volume and increases in sympathetic activity, hydrostatic pressure changes in the lower limbs differ between these conditions. While LBNP application leads to similar increases in vascular resistance in upper and lower limbs (209, 378) during HUT, peripheral vascular responses in the lower limbs appear greater than those in upper limbs. Kitano et al. (209) suggested that the increases in femoral vascular resistance as compared with brachial vascular resistance during HUT could arise due to modulating effects of local mechanisms specifically induced in the lower limbs (e.g., venoatrial reflex and/or myogenic responses) rather than reflex vascular responses induced by unloading of arterial and cardiopulmonary arterial baroreceptors.

2) Both HUT and LBNP induce different regional blood volume responses. For instance, splanchnic volume decreases during LBNP but increases during HUT (358).

Table 4. *Similarities and differences in LBNP and HUT*

	LBNP	HUT
Lower body pooling	−40 mmHg	Yes
Heart rate increase similar at	−50 mmHg	70°
Systolic pressure	↓	↓
Diastolic pressure	↓	↑
Gravity induced	No	Yes
Transmural pressure gradient	↑ throughout lower body subjected to suction	↑ from the heart to the feet
Hydrostatic pressure gradient:		
In lower thorax and within abdomen	No	Yes
Between carotid and aortic baroreceptors	No	Yes
Splanchnic volume (358)	Decreases	Increases

LBNP, lower body negative pressure; HUT, head up tilt. [Modified from Goswami et al. (138).]

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3) HUT induces mild hypocapnia (4, 21, 23, 39) which has been suggested to be associated with ventilation-perfusion differences within the lung. However, several studies indicate that changes in ventilatory CO_2 might not explain the entire effect (23, 187, 337). A study by Ogoh et al. (278) using LBNP has now provided evidence for a possible secondary mechanism which is that orthostatically induced decreases in CBF attenuate the "washout" of brain CO_2, producing hyperpnea following activation of the central chemoreflex. The reductions in CBF, which arise due to postural changes, appear to modify the central respiratory chemoreflex by moving its operating point. A recent study has reported that larger decreases in CBF occur during HUT than during LBNP due to the greater reduction in end-tidal $P_{ET}CO_2$ along with gravitational effects on the brain circulation (29). This is particularly important to consider in cases where LBNP is used as an MRI-comparable alternative to HUT.

In a study conducted to compare the effects of LBNP at −40 mmHg with the effects of central hypovolemia induced by standing on respiratory mechanics (24), two notable differences were observed: functional residual capacity (FRC) and peak intra-abdominal pressure (IAP) both increased with standing but not during LBNP **(FIGURE 6)**.

4) Standing causes a pattern of hydrostatic pressure gradients along the body axis (where free fall is prevented by a $+G_z$ force, either provided by the ground or by a floating platform that induces a footward shift of body fluids) while LBNP in supine persons induces a single pressure step at the seal area. In both models, locations exist where vascular pressure and/or volume does not change with the inception of HUT or LBNP, respectively. In fact, a caudal shift of the venous volume indifference point has been observed with −20 mmHg LBNP (190). No information seems to be available on possible corresponding venous or arterial hydrostatic indifference point shifts with the onset of LBNP, but it is reasonable to expect that hydrostatic indifference point locations may differ between LBNP and HUT, which might explain unequal features of the two paradigms.

5) Nonhypotensive LBNP does not reduce arterial pressure, and arterial baroreceptors do not send orthostatic alert signals to the brain. On the other hand, hypotensive LBNP leads to stimulation of both carotid and aortic receptors, as in a hypovolemic situation. This is in contrast to upright tilt with its dissociation of carotid (decreased frequency) and aortic discharge rate (no change). Such studies highlight how LBNP can be used as a useful tool for investigating specific features of cardiovascular regulation within the area of vascular volume and pressure dynamics.

6) A linear correlation has been reported between tilt angle/LBNP suction application level and the reductions in

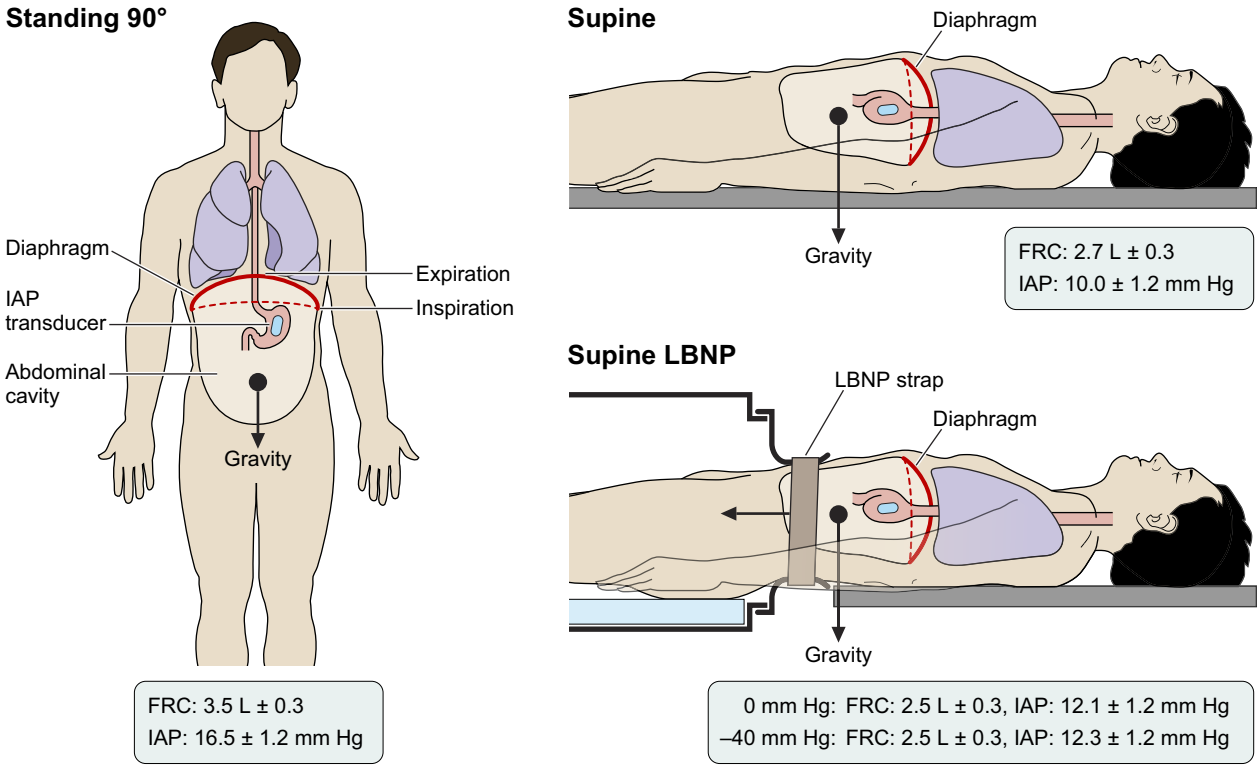


FIGURE 6. Functional implications of gravity and lower body negative pressure (LBNP) on breathing mechanics. FRC, functional residual capacity; IAP, intra-abdominal pressure. [From Blaber et al. (24).]

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central venous pressure, thoracic blood volumes, and atrial diameters in healthy young subjects (95, 273). The extent of decreases in these parameters depends on the tilt angle/level of LBNP suction application as well as the duration of central hypovolemia induced by application of HUT/LBNP (214, 270).

7) Länne et al. (212) reported an increase in muscle and skin capillary filtration coefficient from 54 to 97 $\mu l \cdot min^{-1} \cdot mmHg^{-1} \cdot 100\ g^{-1}$ with −75 to −90 mmHg LBNP, indicating considerable importance of the LBNP-induced fluid filtration in terms of plasma volume control. Due to increased transmural pressure (reduced interstitial pressure) with an application of suction on the lower body (341), LBNP induces outward filtration (5) and a rise in leg volume (358). After transient hemodilution (170), LBNP increases plasma protein concentration as well as hematocrit, signaling hemoconcentration. This observed blood/plasma density increase with LBNP is comparable to transvascular outward fluid shift of HUT that occurs below the location of hydrostatic indifference point (HIP) (125, 149, 166, 190, 250, 295, 297), where capillary pressures rise after assumption to an erect position, and the extent of outward filtration exceeds inward fluid movement in capillaries above the HIP level (i.e., extravasation in the lower body outweighs fluid gain in the upper body). Although the pattern of transvascular fluid exchange is different with LBNP, filtered fluid recirculates to the bloodstream via the lymphatics with considerable delay in both situations (282, 296).

Based on measurements of hematocrit, hemoglobin, plasma proteins, and blood/plasma mass density parameters, the time course of transvascular fluid exchange is different in LBNP compared with HUT. A single-phase nonlinear volume loss was observed with HUT (154, 168), whereas both plasma and blood mass density show a biphasic time course after the onset of moderate (−35 mmHg) LBNP (170).

1. Posture change and the influence of inhibition of the different types of baroreceptors (cardiopulmonary, aortic, and carotid)

Hydrostatic effects on the carotid baroreceptors with central blood volume kept constant have been examined through comparison of upper body posture change from vertical to supine, with and without LBNP (301–303). Pump et al. (301) observed that low-pressure receptors together with aortic pulsation are important for the reductions in MAP and for maintaining heart rate decreases up to 5 min during anti-orthostatic maneuvers (upright seated with legs horizontal to supine). Low- and high-pressure reflexes apparently interact to control MAP and heart rate with the sensitivity of arterial reflexes dependent on the degree of low-pressure baroreflex stimulation.

Stimulation of the low-pressure cardiopulmonary and high-pressure arterial receptors is known to be associated with hormonal regulation of blood pressure and volume through suppression of AVP release (274, 363). Further information on the influence of each type of receptor on AVP was assessed by Pump et al. (302). These researchers used the above-mentioned model of anti-orthostasis (301) to assess whether carotid baroreceptor stimulation was involved in the suppression of AVP. Using the anti-orthostasis model with LBNP (which ensured that left atrial diameter was maintained at the seat level), they concluded that *1*) AVP is indeed suppressed during anti-orthostatic posture change (associated with elevations in carotid sinus pressure, pulse pressure and left atrial diameter), and *2*) AVP suppression is not seen when pulse pressure and left atrial diameter are prevented from increasing. These authors concluded that during moderate posture changes, it is the combination of increases in left atrial pressure and pulse pressure, rather than carotid baroreceptor stimulation per se, which may be responsible for suppression of AVP secretion. The individual contribution of high- and low-pressure receptors in release of vasoactive hormones is particularly important for heart failure patients (who already have higher levels of vasoconstrictor hormones and increased systemic vascular resistance) in whom this information could be used to develop newer sodium and fluid regime therapies (302).

In a subsequent study, Pump et al. (303) investigated whether central blood volume expansion by changing posture from seated to supine position overrides the effects on the heart rate of carotid baroreceptor adaptation and/or buffering of aortic reflexes. Compared were the effects of posture change from seated to supine, seated with neck suction (suction adjusted to reflect hydrostatic pressure increases of posture change in carotid sinus), and supine posture with LBNP application (left atrial diameter unchanged thus preventing low-pressure receptor stimulation). The design of the study was to ensure that both seated with neck suction and supine posture with LBNP would impart the same degree of stimulation of cardiopulmonary low pressure and carotid high pressure receptors while a change in posture from seated to supine and supine with LBNP would lead to similar carotid baroreceptor stimulation, with and without central volume expansion. The authors observed that stimulation of carotid baroreceptors leads to reductions in heart rate during changes in posture from seated to supine but also that the central volume expansion is necessary to maintain this reduction in heart rate (303). In addition, measurements of norepinephrine in this study (303) showed that norepinephrine release into the forearm

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does not depend on carotid baroreceptor stimulation but rather on low-pressure receptor stimulation. The authors also observed that increases in pulse pressure with no concomitant stimulation of the cardiopulmonary receptors are not enough to decrease AVP release. As AVP release is not affected by low-pressure receptor unloading during LBNP (276), these results suggest that combinations of elevations in pulse pressure and left atrial diameter are necessary to attenuate AVP release (303).

B. Advantages of LBNP Versus HUT

When compared with tilt table or stand tests, LBNP has several advantages:

1. LBNP allows for the use of measurement techniques/technologies that are sensitive to motion and/or requires a supine posture (e.g., magnetic resonance imaging, MRI);
2. LBNP allows the maintenance of central hypovolemia in a supine posture without support from the legs, thus allowing investigation of isolated cardiovascular mechanisms involved in blood pressure control without including muscle pump effects;
3. Supine LBNP removes head motion and the subsequent vestibular stimulation, which is known to affect autonomic responses involved in blood pressure regulation;
4. Rapid termination of the central hypovolemic challenge can be executed during LBNP, in contrast to HUT, thereby decreasing the occurrence of vasovagal syncope. In HUT persons, greater time is required to return them to a supine position, which may be associated with the development of presyncope;
5. As LBNP prevents falling during a syncopal event, safety for human subjects is optimized;
6. LBNP provides the capability of determining the total capacity of the cardiovascular system to regulate arterial blood pressure as well as the assessment of compensatory reserve during central hypovolemia by giving investigators the ability to induce a syncopal/presyncopal end point under controlled and relatively safe experimental conditions. Thus the use of LBNP provides a unique capability to identify mechanisms associated with presyncope by capturing physiological data just before the onset of bradycardia, while mechanisms of vasovagal syncope can be investigated by acquiring data after the onset of reduced heart rate and vasoconstriction. In contrast, tilt table or stand tests restrict the subject to a single posture position which may not always induce syncope. Furthermore, the limitation of the tilt table to define orthostatic tolerance could potentially lead to incorrect clinical diagnosis and effectiveness of treatments.
7. LBNP can be rapidly terminated when presyncope develops before the subject experiences loss of consciousness/syncope.

As LBNP, stand, or tilt tests can all potentially lead to vasovagal syncope and/or cardiac arrhythmias, LBNP application in human volunteers should be performed carefully, with adequate monitoring of physiological parameters and in the presence of a medical monitor.

C. Effects of Repeated LBNP and HUT on Hormonal Responses

Roessler et al. (317) investigated the effects of three repeated episodes of 30 min of −55 mmHg LBNP or 70° HUT separated by supine rest of equal duration on plasma hormones [adrenocorticotropic hormone (ACTH), aldosterone, PRA, atrial natriuretic peptide (ANP), and AVP]. They observed that the MAP was elevated with repetitive HUT but not with repetitive LBNP. With respect to the hormones, PRA changes were larger with LBNP than with equivalent levels of HUT, but aldosterone changed more with HUT than with LBNP. Interestingly, the elevations in ACTH correlated with increased aldosterone in both LBNP and HUT ($R^2 = 0.96$). The AVP and ANP responses, however, were similar in the three consecutive LBNP and HUT situations.

Blood pressure is sensed by various receptors, including juxtaglomerular receptors, which respond by activating the renin-aldosterone cascade. A slight drop in mean arterial pressure with LBNP might be compensated for by a greater increase in PRA (44) compared with HUT, where arterial pressure increases, thereby lessening the magnitude of reflex PRA response. The data of Roessler et al. (317) show that with repeated central hypovolemia, plasma aldosterone levels rise (along with ACTH) in a stepwise fashion **(FIGURE 7)**, conceivably increasing the volume-guarding effect of aldosterone. While the strong correlation between aldosterone

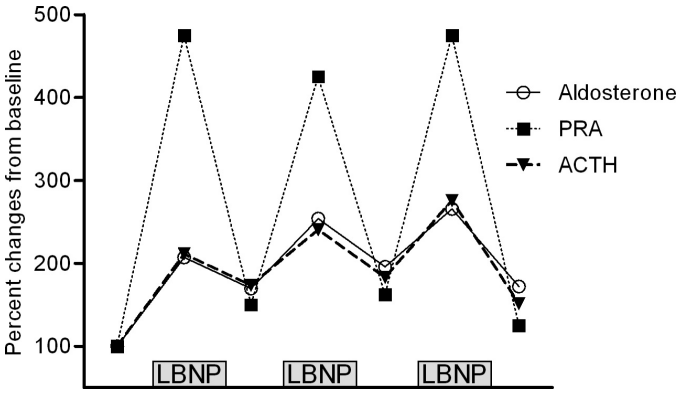


FIGURE 7. Effects of repeated lower body negative pressure (LBNP) on adrenocorticotropic hormone (ACTH), aldosterone, and plasma renin activity (PRA) responses.

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and ACTH during repetitive HUT or LBNP can indicate a predictive relationship, it does not suggest a causal relationship (317).

D. Combining HUT and Graded LBNP

In most studies, development of presyncope has been studied using only graded LBNP (7, 72) or tilt table (205, 327, 390). However, using supine LBNP in individuals with high LBNP tolerance may necessitate the use of very high LBNP suction levels, which may result in misleading physiological responses due to the discomfort of the subject associated with high forces against the footplate/saddle. Therefore, in recent years, the combination of HUT with graded LBNP until subjects reach presyncope has been used to assess tolerance to central hypovolemia (99, 106, 135, 137, 142, 171, 277). Following a supine period, the subject is tilted to ~70° HUT for a specified period [4–5 min in some studies (106, 171) or up to 30 min in others (99)] and then, with the subject still in HUT position, graded LBNP is commenced until presyncope develops. Orthostatic tolerance is defined as the time (in min) from HUT commencement to the development of signs and symptoms of presyncope. The HUT + graded LBNP protocol, in addition, has been used for assessing cardiovascular, autonomic, hormonal, and coagulation responses at presyncope (83, 106, 141, 171). The combined HUT + graded LBNP protocol has also been used to investigate effects of traditional Chinese medicine (TCM) (121), mental challenge (139), and artificial gravity training on the time to presyncope (that is, to assess orthostatic tolerance) (103, 105, 135). Typical cardiovascular responses in males and females to HUT + graded LBNP leading up to presyncope, and how these responses are modified by AG training are shown in **FIGURE 8** (135).

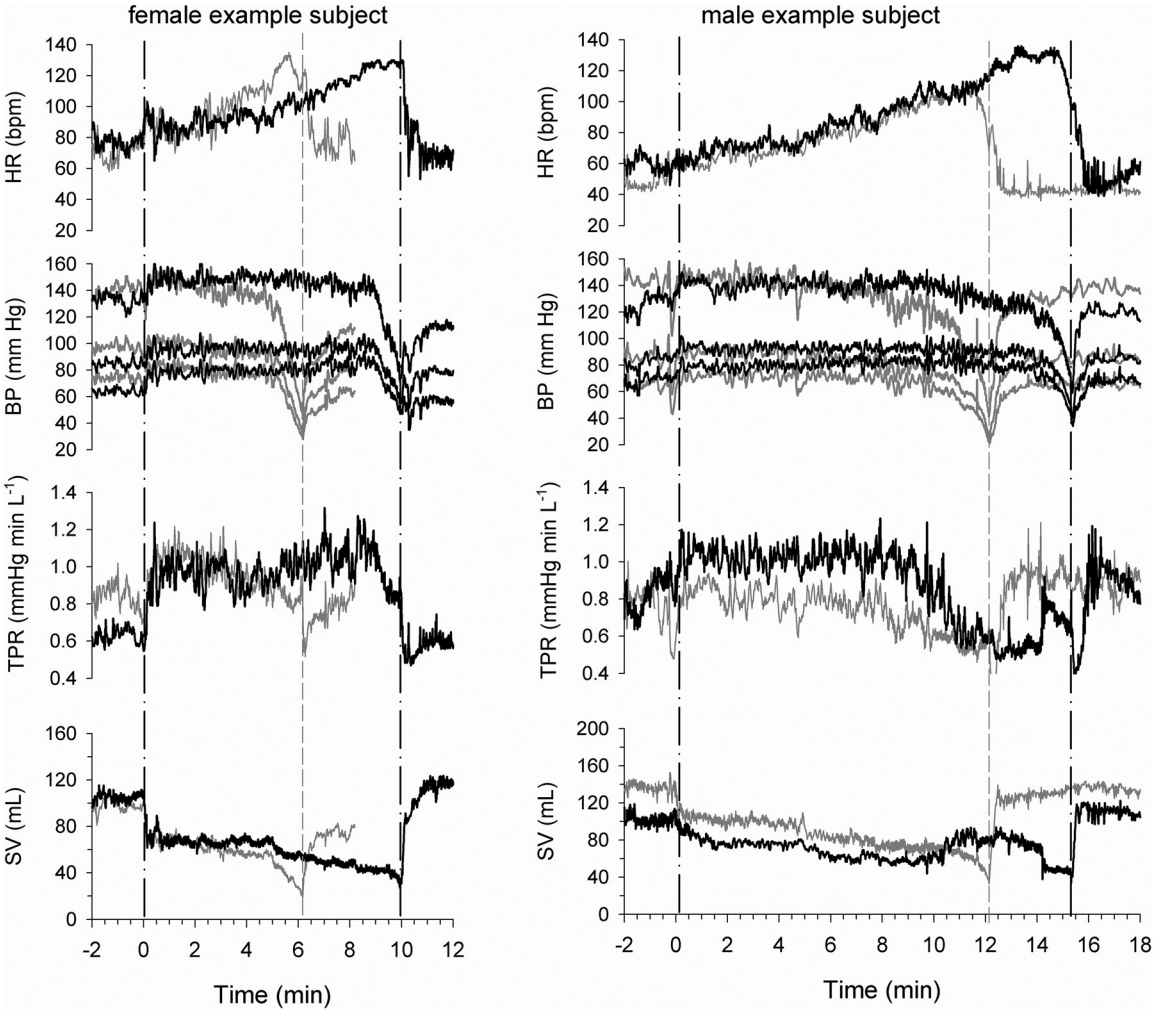


FIGURE 8. Cardiovascular responses across the sexes to head-up tilt with graded lower body negative pressure (LBNP) to presyncope. The physiological responses (HR, heart rate; BP, blood pressure; TPR, total peripheral resistance; SV, stroke volume) in a female subject are shown on the *left*, and those of the male subject are shown on the *right*. Physiological data from the control day are shown in gray, while those during artificial gravity (AG) day are shown in black. The times of presyncope are shown by vertical dashed lines. [From Goswami et al. (135).]

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V. LBNP AS A TOOL FOR STUDYING ADAPTATIONS TO ACUTE AND CHRONIC CONDITIONS OF EXTREME PHYSIOLOGY AND MEDICINE

A. Spaceflight/Microgravity Deconditioning Countermeasure

LBNP sequesters blood in extrathoracic as well as intra- and extravascular compartments and, to an extent, counters the cephalad shift of blood and body fluids seen in hypogravity. Therefore, it can be used to reduce orthostatic intolerance, which follows bed rest or upon return to Earth from the microgravity environment of space flight (126, 220, 385). LBNP has been shown to counter spaceflight deconditioning simulated by prolonged bed rest exposure by providing loading of the musculoskeletal system similar to that occurring on Earth (161).

1. LBNP in space to prevent post-spaceflight orthostatic intolerance

LBNP is employed during spaceflight as a stimulus to provoke neurohumoral stress responses, as well as to protect against orthostatic instability that occurs in two out of three astronauts upon return to Earth (42).

Since LBNP alone provides little protection against cardiovascular deconditioning, combinations with treadmill exercise have been proposed to provide gravity-like conditions to load the musculoskeletal and cardiovascular systems (161). On Earth, this combined procedure not only protects persons subjected to HDT bed rest against orthostatic deconditioning (385), but also improves muscular strength (221), spine function (243), and bone metabolism (347). The effect of LBNP combined with exercise against loss of balance has been reported to be significant in men but not in women (242). It remains to be seen if LBNP combined with exercise will evolve into a routine procedure to prevent post-flight orthostatic imbalance.

Another possible countermeasure for post-flight orthostatic intolerance is fluid loading, with the idea being to increase extracellular fluid volume before a cardiovascular challenge to bolster venous return, cardiac output, and consequently arterial pressure. Crew members have been given salt tablets and water ~2 h before reentry [known as a "salt soak" comprising 1-g salt tablets with ~900 ml of fluid designed to make an isotonic saline drink (33)]. But there is little evidence to suggest that salt supplementation alone is effective against the development of postflight orthostatic intolerance (32) or body fluid stabilization (163). There have been claims that sodium chloride ingestion combined with LBNP protects astronauts against orthostatic intolerance after landing when applied within hours before reentry at the end of a space mission (41, 42). More recent findings suggest that in-flight LBNP combined with fluid loading may indeed provide some protective effect, although the benefits of such combined treatments may not be realized given possible disapproval of such procedures by astronauts (226), rendering the application of such combined treatments questionable for routine in-flight application (43). For details of LBNP experimentation in long-term flights, the reader is referred to the comprehensive work of Levy and Talbot (226).

B. Relation to High-G Tolerance/Adaptation

Given the huge costs of using human centrifuge facilities for identification of $+G_z$ tolerant pilots, enhancement of $+G_z$ tolerance through training, and development of $+G_z$ countermeasures, LBNP may be considered as a potential low-cost surrogate for the study of sustained high-G ($+G_z$) physiology (103, 132). On the basis that both stressors cause head-to-toe blood redistribution, tachycardia, hypotension, and eventual loss of consciousness when subjects are exposed to an excess of either stress, Ludwig et al. (239) performed direct comparisons of tolerance to LBNP and $+G_z$ acceleration. In this investigation of 17 subjects, 3 centrifugation onset rates of $0.01 +G_z \cdot s^{-1}$ (slow), $0.05 +G_z \cdot s^{-1}$ (medium), and $0.2 +G_z \cdot s^{-1}$ (rapid) were compared with 3 LBNP onset rates of 0.067 mmHg$\cdot s^{-1}$ (slow), 0.3 mmHg$\cdot s^{-1}$ (medium), and 2.0 mmHg$\cdot s^{-1}$ (rapid) in a seated LBNP profile. In addition to tolerance measurements, physiological functions of carotid-cardiac baroreflex sensitivity, baroreflex-mediated control of peripheral vascular resistance, and resting vagal tone (R-R interval variation) were noninvasively measured. The primary finding was that tolerance to the fastest LBNP onset rate correlated best with the slow and medium $+G_z$ tolerances, but not with the rapid $+G_z$ onset rate, suggesting that LBNP could be useful as a surrogate for certain $+G_z$ tolerances when lower body decompression is delivered at a rapid rate. The largest and most consistent physiological correlation with both $+G_z$ and LBNP tolerances was carotid-cardiac baroreflex sensitivity, a finding consistent with the observation that the vagally mediated cardiac baroreflex response is a primary contributing mechanism associated with tolerance to hypovolemia (238). As a protective mechanism, the association of cardiac baroreflex responsiveness during rapid onset rates of central hypovolemia to rapid onset rate of seated LBNP and slow-to-moderate onset rates of $+G_z$ may be expected given the rapid (i.e., milliseconds) response of cardiac vagal withdrawal compared with slower sympathetically mediated mechanisms involved with control of cardiac and peripheral vascular responses (96).

Because of the qualitative similarity in physiological stimulus created by acute central hypovolemia, Convertino (56) used LBNP to study adaptations of blood pressure regulatory mechanisms to repeated $+G_z$ exposure. Ten subjects

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underwent three sessions (different days) of exposure to high $+G_z$ acceleration (G-training) on a centrifuge. G-training sessions consisted of acceleration with gradual onset rate of $+G_z$ ($+0.1$ $+G_z \cdot s^{-1}$) for 50 s, ~5 min of recovery, followed by a rapid onset rate of $+G_z$ ($+6.0$ $+G_z \cdot s^{-1}$) to a maximum of $+7.0$ $+G_z$ for 30 s, followed by a series of simulated air combat maneuvers (SACM) consisting of exposure to $+4.5$ to $+5.0$ $+G_z$ for 15 s immediately followed by exposure to $+7.0$ to $+9.0$ $+G_z$ for 15 s. Compared with 10 control subjects who did not undergo $+G_z$ exposure, G-training was associated with reduced vasoconstriction, less decline in stroke volume and cardiac output, and reduced venous pooling in the lower extremities during LBNP. These lower responses to progressive hypovolemia induced by LBNP were associated with greater circulating blood and erythrocyte volumes as well as increased carotid-cardiac baroreflex sensitivity and cardiac β- and vascular α-adrenoreceptor responsiveness. These results demonstrate that the hypovolemic stimulus of repeated exposure to high $+G_z$ is manifested by increased capacities of mechanisms that underlie blood pressure regulation and can be demonstrated with the use of LBNP.

C. LBNP as a Research Tool in Experimental and Clinical Medicine

LBNP has been used in the study of blood pressure regulation (185) and for evaluating how pharmacological therapy affects hemodynamic and neurohormonal responses, including cardiovascular reflexes (89, 90, 93). LBNP can be used with congestive heart failure patients to characterize reduced cardiopulmonary functions (371). In addition, it can potentially be used to separate reflexes involved in blood pressure regulation arising from the aortic arch and the carotid sinus (110) and those reflexes arising from intracardiac regions, which are sensitive to intrachamber volume.

LBNP can also be used to evaluate cardiac function as demonstrated by its application in astronauts who flew on the United States Skylab-4 mission by generating Frank-Starling curves with stepwise reductions in end-diastolic and stroke volumes measured noninvasively with echocardiography (58).

1. LBNP as a tool in neurosensory research

Although the outputs from cardiovascular and vestibular/somatosensory receptors are directed toward different systems with specific physiological functions, neuronal activity from either entity has an influence on the other. Since LBNP causes redistribution of blood along the body's Z-axis without the need of body repositioning, vestibular disturbance, and changed limb position, its application can be used for investigation of regulatory mechanisms specific to cardiovascular homeostasis (30) or confirm cardiovascular stunning such as caused by sympathetic denervation (40). As such, LBNP represents a unique experimental tool for the study of integrative systemic cardiovascular physiology.

LBNP-induced unloading of cardiopulmonary baroreceptors increases protective startle eyeblink responsiveness, indicating that cardiovascular receptors are able to influence adapting response strategies during stressful circumstances (312). In addition, the perception of the subjective visual vertical, mainly based on contributions from visual, vestibular, and proprioceptive systems, is altered by LBNP as applied under microgravitational conditions (237). Altered mechanical input (volume loading) to the central circulation influences somatic reflex patterns: LBNP release can suddenly increase blood flow to the lungs, and venous return may rise as much as one liter of blood under such conditions (235) and thereby stimulate pulmonary J receptors (48, 284). This procedure can transiently diminish the amplitude of the tendon (Hoffman) reflex and increases tidal volume (3). These findings suggest the possibility of a LBNP influence of cardiopulmonary receptor input upon neuronal somatic circuits.

LBNP has been an important tool for the investigation of neural control of blood pressure, particularly in the examination of the forebrain architecture associated with baroreflex-mediated sympathetic activation during simulated hemorrhage. A magnetic resonance imaging compatible LBNP device was developed by Kimmerly et al. (206) to induce graded venous pooling and, hence, baroreceptor unloading in otherwise healthy participants. Through progressive central blood volume reduction, Kimmerly et al. (208) went on to demonstrate involvement of the central cortical network in cardiovascular control. They reported that forebrain (frontal and parietal) and left cerebellar activities were increased during conditions of central hypovolemia caused by application of LBNP in humans. Other brain structures, such as the anterior insula, the amygdala, or the mediodorsal thalamic nucleus, were less activated, an effect that was observed with unchanged arterial mean or pulse pressures during LBNP. They concluded that tonic suppression of sympathoexcitatory brain activity may be removed by LBNP-induced baroreceptor unloading. LBNP also activates central structures such as the cingulate cortex and insula, and sensory stimulation inhibits sympathetic activity during LBNP-induced baroreceptor unloading, suggesting baroreceptor input-driven effects on the processing of somatosensory afferents within the forebrain. These studies highlight that during baroreceptor unloading there are strong associations between heart rate and elevated right superior insula and the dorsal anterior cingulate activation.

LBNP induces fluid shifts into the legs while HDT causes headward movement of blood from the lower half of the

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body. Combining the two stressors to equate intensities can neutralize their effects on pulmonary and cardiovascular baroreceptors, and thereby suppress respective reflex (215) and hormonal, as well as hemodynamic and fluid shift effects (175). LBNP combined with HDT (169) can be used to study, for example, vestibular responses to altered body positioning that are undisturbed by parallel influences from cardiopulmonary and vascular baroreceptors. Complete "nomograms" of referring stimulus combinations, however, still need to be developed.

2. Simulating hemorrhage/hemorrhagic shock

Exsanguination is a leading cause of death in both civilian and military trauma, and hence, there is the necessity to develop a model that mimics the physiology of human hemorrhage. Such a model is critical to understanding mechanisms that underlie the failure to compensate for blood loss that can lead to circulatory shock (i.e., inadequate tissue oxygenation) and death. Unlike animal models of hemorrhage, LBNP provides a noninvasive method capable of inducing progressive reductions in central blood volume in conscious humans that can model aspects of the hemodynamics associated with bleeding. Using LBNP circumvents ethical and safety considerations of removing large volumes of blood from human subjects, including a safe procedure in which symptoms of progression to cardiovascular collapse can be immediately resolved by terminating the decompression.

Initial comparisons of LBNP to hemorrhage depended on the collection of data generated from independent laboratory experiments in which human subjects were either exposed to LBNP or blood volume withdrawal. Such comparisons yielded ranges of estimations that −10 to −20 mmHg LBNP approximated 400 to 500 ml (~10%) blood loss, −20 to −40 mmHg LBNP approximated 550 to 1,000 ml (10–20%) blood loss, and greater than or equal to −40 mmHg LBNP approximated >1,000 ml (>20%) blood loss (78). Further indirect support of the notion that LBNP can accurately mimic the hemodynamic responses to human hemorrhage was reflected by validation of a computational platform developed from human hemorrhage data (356). The only experiment that involved a direct comparison of LBNP to hemorrhage demonstrated similar stimulus-response relationships between reduced central venous pressure and elevated sympathetic nerve activity (310). However, this study was limited to minimal nonhypotensive levels of hemorrhage (450 ml) and LBNP (−15 mmHg).

Recently, the first study designed to obtain direct comparisons of hypotensive blood volume loss to reductions in central blood volume induced by LBNP was conducted using non-human primates (177). The experimental design included matching central venous pressure at baseline rest and four levels of relative blood loss (6.25, 12.5, 18.75, 25%) to four levels of LBNP. These experiments eloquently demonstrated that hemodynamic responses to actual progressive hemorrhage that resulted in significant hypotension and hemodynamic decompensation in one-third of the animals was accurately reproduced by LBNP. More striking was the observation that human LBNP responses were identical to the primate responses to both LBNP and hemorrhage **(FIGURE 9*A*)** (177). These data allowed for the first time a direct comparison of a wide range of clinically significant volume of blood loss (25%) from an actual controlled hemorrhage to be translated to LBNP in humans such that −30, −60, and −90 mmHg LBNP is equivalent to ~450, 1,000, and 1,600 ml blood loss in a human with 70 kg body weight **(FIGURE 9*B*)**. Although not as systematically designed, subsequent investigations in human subjects have demonstrated that stimulus-response relationships between hemodynamic responses and progressive reductions in central venous pressure are similar between LBNP and hemorrhage (192, 372). As such, LBNP has repeatedly proven to accurately simulate physiological responses to actual blood

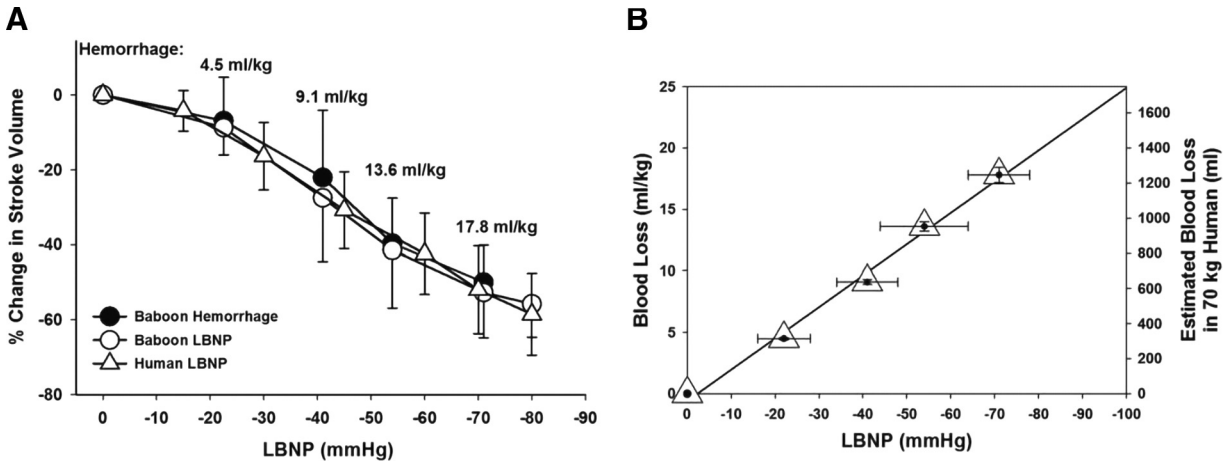


FIGURE 9. Human lower body negative pressure (LBNP) responses are identical to the primate responses to both LBNP and hemorrhage. [From Hinojosa-Laborde et al. (177).]

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loss, and this has been instrumental for advancing knowledge about the human physiology underlying hemorrhage and shock (331).

D. Acute and Chronic Exercise Responses

In the late 1980s, Eiken and colleagues (97, 98) performed experiments using supine LBNP to modify lower limb hemodynamics to investigate muscle blood flow and exercise performance. Their hypotheses were based on evidence (presented in sect. IV*A*) that LBNP produces cardiovascular responses similar to the transition to the upright position. Their initial study showed that supine leg exercise at −50 mmHg LBNP induced an increase in stroke volume that was consistent with that obtained in the upright body position (98). In a follow-up study, eight subjects were asked to perform leg exercise in the supine position with and without exposure to −50 mmHg LBNP and a further nine subjects performed the same exercise, with the addition of supine and upright exercise without −50 mmHg LBNP (97). Although LBNP was found to reduce cardiac output over a wide range of work loads, work performance was improved in the LBNP compared with the supine control condition. This was further evidence that LBNP provided conditions that were closer to supine exercise (98). One of the benefits of the application of LBNP would be the improvement of muscle perfusion via changes in transmural pressure. This not only explains improved leg muscle performance in supine LBNP but also when upright, with peripheral circulation in this case contributing to greater improvements than cardiac output.

The ability of LBNP to not only establish cardiovascular responses related to those of upright posture but to also induce muscle blood flow and performance to that close to upright posture (see sect. IV*A*) plays an important role in our discussion of spaceflight exercise countermeasures. Interest in the impact of exercise training on LBNP tolerance may have evolved from anectdotal observations during the Apollo space program. During the final mission of the first United States space station (Skylab), it was reported that application of LBNP during flight produced higher heart rates and episodes of presyncope in the scientist and mission pilot who had higher maximal oxygen uptake ($\dot{V}O_{2max}$) than the mission commander who had a lower aerobic capacity (50). Subsequently, experiments conducted at the Lovelace Foundation corrobated the Skylab observations when lower LBNP tolerance was reported in five runners with an average $\dot{V}O_{2max}$ of 50 ml·min^{-1}·kg^{-1} compared with five nonrunners with an average $\dot{V}O_{2max}$ of 34 ml·min^{-1}·kg^{-1} (240). Additional studies revealed evidence that individuals who habitually participate in endurance exercise training have demonstrated lower LBNP tolerance which was associated with mechanisms such as depressed carotid and aortic baroreceptor reflex responsiveness, high cardiac vagal tone, and greater venous compliance in the lower extremities (50, 306). These observations were interpreted to indicate that high aerobic fitness and endurance training may reduce orthostatic tolerance. The primary limitation to these studies was that the experimental designs only included cross-sectional comparisons of individuals with high versus low aerobic capacities rather than longitudinal comparisons of LBNP tolerance in the same individual before and after exercise training.

With the emergence of longitudinal experimental designs, more evidence was accumulated that supported the notion that exercise training designed to increase aerobic capacity can increase tolerance to central hypovolemia induced by LBNP (50, 53). Adaptation of several physiological mechanisms with chronic exercise training could be associated with enhanced blood pressure regulation. For instance, longitudinal exercise training failed to demonstrate changes in leg venous compliance or cardiac and vascular baroreflex responsiveness (53). However, a fundamental adaptation to chronic exercise training is the expansion of blood volume (52). Increased circulating blood volume will attenuate the heart rate and vasoconstrictor responses during LBNP (362). As such, increased circulating blood volume following exercise training should increase cardiac filling and stroke volume (i.e., the Frank-Starling effect) which should require smaller baroreflex-mediated tachycardia and vasoconstriction as an appropriate corrective response to maintain cardiac output compared with a smaller stroke volume in the pretraining state. This concept is supported by the observation that the lower tachycardia observed during LBNP after exercise training that increased $\dot{V}O_{2max}$ is compensated for by higher stroke volume, thus maintaining cardiac output at or above levels before training (63). Lower heart rate and vasoconstrictor response following exercise training, therefore, reflects a greater reserve for increasing tachycardia and vascular resistance during LBNP compared with the pretraining state and thus increase tolerance to central hypovolemia.

Data generated from longitudinal exercise training studies still provide a challenge to find an explanation for the observations from cross-sectional comparisons that athletes who habitually engage in endurance physical activity (which maintains relatively high $\dot{V}O_{2max}$) tend to display a predisposition to syncope. In an effort to address this dilemma, an alternative hypothesis emerged that described evidence that athletes have structural changes in their cardiovascular system that, although beneficial during exercise, can lead to an excessively large decrease in stroke volume during progressive central hypovolemia such as that induced by LBNP (see sect. III*G4* for details).

It should be emphasized that syncope in athletes occurs in very few highly endurance-trained individuals. Since there is an important genetic component to being an elite athlete,

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it is not unreasonable to speculate that their steeper Frank-Starling curve may be an innate characteristic, thus predisposing them to syncope. The phenomenon of syncope in the elite athlete most likely represents a mechanical change in the cardiovascular system that benefits the athlete during exercise and should not be interpreted as a clinical abnormality. In contrast, the evidence from longitudinal studies using LBNP to test physiological adaptations supports the notion that exercise training provides the benefits of increasing aerobic fitness and tolerance to reduced central blood volume.

E. Studying Visual Impairment Due to Intracranial Pressure

It has been reported that spending 6 mo in space leads to development of microgravity ocular syndrome (MOS) also known as visual impairment and intracranial pressure (VIIP) in half of the astronauts (detailed in Ref. 245). This spaceflight associated neuro-ocular syndrome (SANS) (401), consistent with an increase in intracranial and periorbital cerebrospinal fluid (CSF) during spaceflight, is manifested as reduced near visual acuity, optic disk edema, cotton wool spots, retinal nerve fiber layer thickening, optic nerve sheath diameter (ONSD) increases, and eye globe flattening. Since it was first recognized as a problem in astronauts (249), visual impairment has been an issue of high priority in defining health risk factors associated with prolonged spaceflight. The primary hypothesis for such impairment put forward in the position paper published by NASA (2) was that elevated intracranial pressure (ICP) was the proximal cause with a number of environmental and physiological conditions in spaceflight cited as possible underlying factors, including exercise training and elevated ambient CO_2.

Macias et al. (244) demonstrated that elevated ICP could be induced in ground-based studies through acute exposure to HDT. These investigators also reported that LBNP application attenuates the HDT-induced increases in ICP. Recently, Marshall-Goebel et al. (254) performed an HDT study in nine healthy males, with and without LBNP, in which the effects of headward fluid shifts on intracranial CSF volume and ONSD (as a measure of periorbital CSF changes) were assessed. LBNP was applied, as its application has previously been shown to lead to reductions in central venous pressure and CBF as well as intraocular pressure (IOP) (244). They observed that gravitational vectors appear to play an important role in the distribution of CSF, especially in the periorbital ON sheath. The authors suggested that the LBNP-induced effects on the ONSD during HDT appear to arise due to decreases in central venous pressure and improved cerebral drainage. Clearly, the impact on reducing CVP, CBF, and IOP makes LBNP a candidate for the study of mechanisms underlying the development of VIIP.

1. The potential role of LBNP in counteracting space-induced VIIP

Based on the evidence from HDT studies outlined in the previous section, it appears that LBNP may be a promising therapeutic countermeasure for providing footward fluid shifts on board the International Space Station (ISS) (245). In 2015, an experiment with the chibis suit (pants in which LBNP can be applied) was carried out (351) on the ISS to investigate if LBNP can alleviate ocular and visual problems during weightlessness. However, at the time of publication of this review, these results have not been published.

F. Application of LBNP for Physiological Measurements and Clinical Diagnosis

High intrasubject reproducibility in physiological responses to various levels of LBNP provides confidence in the application of LBNP as a potential diagnostic tool of patient autonomic and cardiovascular functions: LBNP-induced central hypovolemia could lead to abnormal responses such as exaggerated venous blood pooling or attenuated tachycardia, which could suggest altered autonomic or physiological functions.

Modified LBNP chambers that allow left side rotation of the subject for the use of echocardiography can be used to assess cardiac function (55). The relationship derived by plotting left ventricular end-diastolic volume against stroke volume changes induced by graded LBNP protocol up to -50 mmHg could be a noninvasive technique for obtaining a Frank-Starling curve. Indeed, such dynamic measurements of cardiac function in patients are even more informative than a characteristic resting echocardiogram (55).

In addition to being used to measure low-pressure baroreflex function, LBNP can also provide information on baroreceptor-mediated reflexes as well as their interactions with other cardiovascular reflexes. The protocol for this assessment involves measuring baseline values of venous and arterial blood pressure and heart rate for a subject in the supine position to provide a baseline reference state. In the supine position, MAP reflects the pressure at both carotid and arterial baroreceptors (55). Arterial and venous pressure baroreceptors can be stimulated with a steady infusion of an adrenergic agonist such as phenylephrine into an antecubital vein. This infusion is given with a target to increase MAP by 15 mmHg. LBNP is subsequently applied until CVP returns to the baseline level existing before phenylephrine infusion. In this way, LBNP is used to remove the stimulus to low-pressure venous baroreceptors so that the resulting change in heart rate is due to the reflex response to stimulation of the arterial baroreceptors. Following 1 min of LBNP application, positive pressure is applied at the position of the carotid baroreceptors using a neck pressure chamber (55). This positive pressure is applied to return

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mean carotid sinus transmural pressure to baseline. Clamping CVP via LBNP and carotid pressure via positive neck pressure at baseline levels is designed to remove phenylephrine-induced loading of cardiopulmonary and carotid baroreceptors. The result is that the aortic baroreceptor influence on cardiac baroreflex response can be isolated. The responsiveness of heart rate to baroreflex control by combined carotid and aortic baroreceptor action and by aortic baroreceptor action alone can then be assessed by calculating the ratio of the difference in heart rate to mean arterial pressure (ΔHR/ΔMAP) between baseline and phenylephrine responses with LBNP and LBNP plus neck pressure (55).

1. LBNP as a tool for investigating clinical and medical diagnostic technologies: some methodological considerations

Prior to any clinical use, physiological methods and technologies must be validated. Given its proven ability to simulate hemorrhage and to also reliably generate reproducible physiological responses, LBNP can be a powerful tool in this process. For example, the sequence method of arterial baroreflex assessment relies on the detection of three or more beats in which RR interval (RRI) changes in the same direction as the blood pressure (i.e., RRI and BP both increase or decrease together). A linear regression is then applied to the data, and the slope represents the "spontaneous" baroreflex gain (ms/mmHg). When the sequence method was first proposed as a technique for arterial baroreflex assessment (18, 288), a one-beat delay between values of SBP change and RRI response was used since this led to the greatest number of sequences. However, a subsequent study (26) showed that in resting human participants there was no delay. The discrepancy was thought to be related to the lag between baroreceptor activation and neural stimulation at the sinoatrial (SA) node: ~775 ms (299). At higher heart rates (short RRI), the signal would reach the SA node too late to affect the current beat. This hypothesis was verified through the use of LBNP to experimentally alter heart rate in a progressive pattern (26). The transition from no delay to a one-beat delay was found to be close to 850 ms.

The use of LBNP in combination with both microneurography of MSNA and strain-gauge plethysmography has been used as methods to assess vasoconstrictor function during venous pooling. However, neither is practical in a clinical setting. Since near-infrared spectroscopy (NIRS) is capable of measuring both oxygenated and deoxygenated blood volumes in tissues, Hachiya et al. (151) demonstrated that NIRS could be used as a surrogate method to assess blood flow in the legs. With the use of a series of LBNP experiments designed to systematically induce vasoconstriction, deoxygenated hemoglobin (deoxy-Hb) and oxygenated hemoglobin (oxy-Hb) were compared with leg volume changes (stain gauge) and MSNA. As long as muscle activity was minimal, both NIRS measures were found to be highly correlated with their invasive counterparts, thus defining a noninvasive clinical methodology for vasoconstrictor function.

Tavakolian et al. (360) also demonstrated that during graded LBNP, seismocardiograph (SCG) signals (cardiac vibrations measured via an accelerometer placed on the xiphoid processes) could successfully detect changes in cardiac function correlated with central blood volume loss. The two most connected measurements were left ventricular ejection time (LVET) ($r = 0.89$, $n = 32$) and pre-ejection period as a function of LVET ($r = 0.90$, $n = 32$). Later the same research group showed that the causal relationship between continuous MAP and RRI heart rate was altered during LBNP (376). Using LBNP, they demonstrated that early changes in blood pressure regulation during central hypovolemia could be detected with their analysis and that traditional baroreflex estimates using heart rate and SBP were ineffectual. These studies, and others cited above in this review, demonstrate the versatility of LBNP in being able to investigate important clinical and medical diagnostic and monitoring technologies.

G. Mathematical Modeling of Physiological Control Systems as Related to LBNP

A powerful tool for synthesizing information about the interaction of complex physiological systems is mathematical modeling. Applications of theoretical mathematical areas such as optimization theory, control theory, sensitivity analysis, and neural network analysis have been incorporated into physiologically based models to study the intricate interaction of physiological control mechanisms. In particular, the complexity of human cardiovascular physiology in response to various experimentally controlled scenarios involving LBNP can be modeled allowing for the examination of factors influencing tolerance to LBNP and for comparing responses to varying seal locations, and HUT. Comprehensive overviews of such modeling activities have been provided by several investigators (13, 338, 389). Information from modeling can lead to a better understanding of cardiovascular responses to blood volume changes in general and provide insight into key clinical problems related to the effects of central hypovolemia (14).

Early modeling of LBNP-induced responses includes the study by Croston and Fitzjerrel (81). This mathematical modeling study of dynamic cardiovascular control responses was motivated by the need to simulate responses to cardiovascular experiments related to Skylab space missions and incorporating various scenarios including application of LBNP challenge.

Sud et al. (354) applied a model to investigate the influence of LBNP on regional blood flows. The model incorporated

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Table 5. *Representative modeling studies, modeling approaches, and applications*

Investigators	Mathematical Modeling Focus	What Simulated	Compartment Types	Heart	Controls
Croston and Fitzjerrell (81)	Dynamic cardiovascular control responses to various stresses	LBNP and relation to microgravity during Skylab missions	28 cardiac and vascular segments Nonlinear vascular compliance	Pulsatile flow	Arterial baroreceptors
Melchior et al. (260)	LBNP challenge under various scenarios including variations in arterial baroreflex sensitivity or reduced leg blood capacity	Graded LBNP up to −40 mmHg	9 compartments Nonlinear vascular compliance		Arterial and cardiopulmonary baroreflex control of peripheral resistance and heart rate
Peterson et al. (298)	Cardiac function and intrathoracic and intravascular pressure	Tilt LBNP	Arterial and venous vascular compartments Nonlinear vascular compliance	Pulsatile flow	
Heldt et al. (164)	Comparison of HUT and LBNP effects with a focus on orthostatic intolerance [ordinary differential equation (ODE) model]	Tilt LBNP	12 cardiac and vascular segments Nonlinear vascular compliance Capillary filtration	Pulsatile flow	Arterial baroreceptors control of systemic resistance, heart rate and filling capacitances, unstressed volume Cardiopulmonary receptors control of systemic resistance and unstressed volume
Summers and co-workers (355, 356)	Global cardiovascular responses to hypovolemia including responses to hemorrhage and especially related to microgravity (computational computer model)	+Gz, hemorrhage LBNP	Based on Guyton/Coleman/Summers model of global cardiovascular system function Nonlinear vascular compliance Capillary filtration	Pulsatile flow	Arterial baroreceptors control of systemic resistance, heart rate and stroke volume Control of venous tone Local vascular regulation
Etter et al. (102)	How graded LBNP affects cardiovascular parameters Focus on influence of caplillary filtration	Tilt + graded LBNP until presyncope	7 compartments Nonlinear vascular compliance Capillary filtration	Pulsatile flow	Arterial baroreceptors control of systemic resistance, heart rate and stroke volume Local vascular regulation
Hao et al. (159)	How hypovolemia impacts time to presyncope	LBNP simulation	7 submodels Nonlinear vascular compliance	Pulsatile flow	Arterial and cardiopulmonary baroreflex regulation of systemic resistance, heart rate, and venous tone
Kappel et al. (201)	LBNP application at two seal locations (upper abdomen and iliac crest) (ODE model)	Impact of LBNP seal location on controls including mobilizing of venous unstressed volume	6 vascular pulmonary vasculature and upper/lower body systemic arterial and venous vasculature	Nonpulsatile flow	Arterial baroreceptor control of systemic resistance, venous unstressed volume, and heart rate

Continued

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Table 5.—*Continued*

Investigators	Mathematical Modeling Focus	What Simulated	Compartment Types	Heart	Controls
Sud et al. (354)	LBNP influence on regional blood flow (finite element model)	LBNP	Arterial, capillary, and venous vascular segments	Pulsatile flow	Model of factors influencing flow relationships reflecting fluid distribution under LBNP
Olufsen et al. (283)	Standing effects on systemic and cerebral blood flow and pressure (ODE model)	Stand test	11 compartments reflecting heart, arterial, and venous vascular segments	Pulsatile flow	Arterial baroreceptor control of systemic resistance, vascular compliance, heart rate, and contractility Cerebral autoregulation

LBNP, lower body negative pressure; HUT, head up tilt.

a finite-element computational approach to study pressures and flow velocities reflecting changes during simulated LBNP in various vascular segments. Simulations indicated that the consequences of LBNP application are reflected primarily in lower body vascular segments.

A series of mathematical modeling studies by Melchior and co-workers (259, 260) were employed to study LBNP challenge under various scenarios including variations in arterial baroreflex sensitivity or reduced leg blood capacity (LBC). The developed model simulated responses to LBNP from 0 to −40 mmHg which were compared with data involving LBNP during autonomic blockade and during use of leg stockings to reduce LBC.

A representative summary of modeling studies, which reflects the variety of modeling approaches and applications, is provided in **TABLE 5**.

H. LBNP as a Teaching Tool in Physiology

In laboratories where an LBNP apparatus and medical supervision are available, LBNP can be used as a teaching tool for simulating central hypovolemia in physiology and medical school curricula (61, 138). Such a learning activity where students can be directly involved in designing and carrying out LBNP protocols can be used to explore the complex nature of cardiovascular changes that occur during central hypovolemia and "simulated orthostasis." It also can illustrate that LBNP-induced central hypovolemia leads to hemodynamic and neurohormonal responses, which vary in different individuals and from one environment to another for the same individual (60, 61, 138). This activity also can reinforce the need to identify possible confounding variables that can potentially affect cardiovascular and hormonal responses associated with acute reductions in central blood volume, and to emphasize that those factors should be considered when defining participant inclusion and exclusion criteria in LBNP studies. In addition, such learning activity can illustrate concepts related to the design of LBNP protocols and the identification of appropriate laboratory conditions related to specific protocols (see Ref. 138). During LBNP classwork, students can learn to *1*) highlight the different uses and applications of this noninvasive physiological tool, *2*) identify the variability in the LBNP experimental protocols used, *3*) recognize the individual variability in physiological compensatory response to central hypovolemia, and *4*) understand how cardiovascular and neurohormonal responses vary during and following different levels and durations of LBNP exposure. Ultimately, such learning activities will also provide a basis for the students to carry out future independent research in both the basic sciences and medical sciences.

1. Teaching systemic integrative physiology using LBNP as a "dry laboratory" activity

LBNP can be used to teach students the physiological effects of central hypovolemia in the absence of the LBNP box and/or a medical monitor, by having students carry out an alternative activity/"dry lab"/exercise using published LBNP papers (61, 138) with which students can plan an experiment, and then speculate on the results. The teacher can conclude this alternative activity by showing the students data reflecting LBNP-induced hemodynamic responses in subjects not developing presyncope and then showing the students data on hemodynamic responses from subjects depicting cardiovascular collapse, with development of presyncopal signs and symptoms.

2. Designing teaching seminars incorporating LBNP applications

Based on the classical paper of Foux et al. published more than 25 yr ago (113), in which the effects of LBNP on

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fluid shifts and homeostatic mechanisms were discussed, teaching seminars for graduate students can be designed (see Ref. 133). During such seminars, students are provided the paper by Foux et al. (113), in which it as reported that application of LBNP at −30 mmHg for a half hour leads to *1*) a shift of blood from the circulatory system via translocation of fluids from the circulation to the tissues, totaling up to 1–2 kg; *2*) a fluid shift between tissue and vascular spaces that is "proportional to the (LBNP) pressure gradient and exponential with time"; and *3*) fluid shifts that occur in two phases: a rapid change due to the LBNP application representing the shift of fluid between vascular regions and a slower shift due to vascular-interstitial fluid exchange. The students will first examine the findings of Foux et al. (113) and then compare these findings with more recent research (170), which has reported that blood volume changes during and after LBNP follow a more complex time course. Students then explore how rapid blood volume changes are caused by an altered balance of Starling forces, which depend on hydrostatic as well as externally applied forces, such as LBNP application. These observations can illustrate how simpler approximations, corresponding to first-order kinetics (113), can be extended to more complex modeling which might be needed to capture, for example, the biphasic filtration and blood volume responses described in earlier sections of this paper (170, 172). Such teaching seminars, by incorporating (and comparing) classic results with more recent scientific research related to LBNP protocol design and experimentation, will help students appreciate how research proceeds and how scientific ideas evolve which will lead to greater flexibility in thinking and open-mindedness when carrying out scientific research. Furthermore, learning about the importance of comprehensive protocols leading to carefully controlled studies can help the future researcher reduce confounding variables and provide a basis for an optimal analysis and elucidation of the physiological responses that are being investigated (outlined in **TABLE 2**).

In addition, such seminars can be adapted to provide the opportunity to develop greater collaboration between medical and life science researchers on the one hand and technical and mathematical researchers on the other which can lead to new opportunities for the implementation and exploitation of emerging technologies in data acquisition, instrumentation, and advanced mathematical modeling (13, 14, 102). Introducing students to such an integrative and interdisciplinary approach will promote innovative research incorporating multiple perspectives. The seminars can thus provide a basis for developing mutual understanding between diverse disciplines that can lead to novel insights and new approaches to problem-solving.

VI. CONCLUSIONS AND RECOMMENDATIONS

LBNP is an experimental platform for studying the physiology of systemic hypovolemia. Paramount to this methodology is the appreciation of how LBNP can be used to explore the complex nature of physiological responses to specific stresses. Understanding the responses induced by various protocols of LBNP can be used to illuminate key physiological mechanisms and clinical conditions. LBNP-induced hemodynamic and neurohormonal responses and tolerance vary from one individual to another but are reproducible in the same individual across time (62, 138). It is necessary to consider numerous confounding variables that could affect LBNP-induced responses (**TABLE 3**) when defining criteria for experiment design and selection of participants.

As each specific research aim necessitates the usage of different LBNP applications, details, and variations of the LBNP protocol have to be carefully considered. Therefore, the categorization of protocols based on LBNP needs to incorporate and clearly specify such factors (e.g., **TABLE 2**).

Given the diverse applications of LBNP presented in this review, it is to be expected that new and innovative applications of LBNP will be developed to explore the complex physiological mechanisms that underline health and disease.

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DISCLOSURES

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